TECHNICAL REPORT
R-58
First version: Jun 2020
Last revision: Nov 2020

Causal Effect Identifiability under Partial-Observability

Sanghack Lee [1] **Elias Bareinboim** [1]

Abstract

Causal effect identifiability is concerned with establishing the effect of intervening on a set of variables on another set of variables from observational or interventional distributions under causal assumptions that are usually encoded in the form of a causal graph. Most of the results of this literature implicitly assume that every variable modeled in the graph is measured in the available distributions. In practice, however, the data collections of the different studies considered do not measure the same variables, consistently. In this paper, we study the causal effect identifiability problem when the available distributions encompass different sets of variables, which we refer to as identification under partial-observability. We study a number of properties of the factors that comprise a causal effect under various levels of abstraction, and then characterize the relationship between them with respect to their status relative to the identification of a targeted intervention. We establish a sufficient graphical criterion for determining whether the effects are identifiable from partially-observed distributions. Finally, building on these graphical properties, we develop an algorithm that returns a formula for a causal effect in terms of the available distributions.

1. Introduction

One of the central goals in data sciences (the health and the social sciences), artificial intelligence, and machine learning is to discover cause and effect relationships. If the scientific study is performed appropriately, and causal relations are eventually discovered, the corresponding effects are more likely to hold under a broader set of conditions. Causal relations are usually more stable and generalizable across disparate conditions (Pearl, 2000; Pearl & Mackenzie, 2018).

[1]Department of Computer Science, Columbia University, New York, NY 10027, USA. Correspondence to: Sanghack Lee <sanghacklee@cs.columbia.edu>.

Proceedings of the 37th *International Conference on Machine Learning*, Online, PMLR 119, 2020. Copyright 2020 by the author(s).

Causal inference provides a collection of principles and tools to help understand the conditions under which these extrapolations can take place (Pearl, 2000; Spirtes et al., 2001; Bareinboim & Pearl, 2016). For instance, a scientist may be able to use an observational study to infer the effect of a new intervention by leveraging knowledge encoded in its causal model (Pearl, 1995; Tian & Pearl, 2002; Tian, 2002; Shpitser & Pearl, 2006a; Huang & Valtorta, 2006b). Causal effects can also be inferred across a broad range of conditions, including across disparate populations (Bareinboim & Pearl, 2014; Lee et al., 2020), under selection bias (Bareinboim & Pearl, 2012b; Correa & Bareinboim, 2017), missing data (Mohan & Pearl, 2014), and in the absence of the causal graph (Jaber et al., 2019), to cite a few. Further, recent advances in causal inference lead to algorithmic solutions to combine data collected under multiple, disparate regimes (observational and interventional) to identify a causal effect (Lee et al., 2019). Despite all the generality and power provided by these results, by and large, they implicitly assume that every modeled variable is consistently available across the different data collections. Since each study is usually designed to fulfill its own objectives, datasets across studies tend to measure different sets of variables (i.e., the datasets have different columns). As a consequence, relevant data available to answer a question about a specific effect may be not usable in another, possibly very related study.

For concreteness, consider the setting where a researcher aims to understand the effect of physical exercise (X) on stroke (Y), written as $P_x(y)$, and is then analyzing two related datasets. The first is based on an experimental study estimating the effect of physical exercise itself (X) on blood pressure (C), collected from different age groups (A); commonly written as $P_X(A, C)$. The second dataset is based on an observational study about the association among body mass index (BMI, or B), blood pressure (C), and stroke (Y), i.e., $P(B, C, Y)$. The causal graphs representing these two studies are shown in Fig. 1a and 1b, respectively. After conducting standard causal analysis, the researcher realizes that $P_x(y)$, the effect of interest, cannot be inferred from the two datasets, separately. She then creates a common representation of their union, which is summarized in the causal graph shown in Fig. 1c. Unfortunately, the identification algorithms available today assume *full-observability*, which means that graphs and datasets

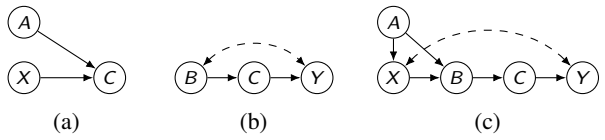


Figure 1. (a, b) Causal graphs representing experimental ($P_X(A, C)$) and observational studies ($P(B, C, Y)$), respectively. (c) The causal graph representing the union of both studies.

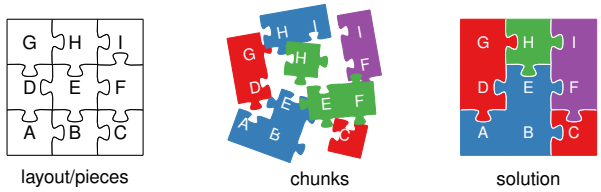


Figure 2. A high-level abstraction of our problem as a jigsaw puzzle. A causal query $P_{\mathbf{x}}(\mathbf{y})$ is a puzzle with how its pieces (factors) should be laid out. Each of the available distributions (represented as different colors) provides *chunks* of information, where a chunk is either a piece or the combination of multiple pieces. The solution for the puzzle is then putting the subset of chunks together.

defined over different sets of variables cannot be taken as input. On the other hand, the effect $P_x(Y)$ is inferable by the careful combination of these studies through the expression $\sum_{a,c} P_x(a, c) \sum_b P(Y|b, c)P(b)$. The first factor can be computed from the experimental study, while the other two can be obtained from the observational study.

Our goal in this paper is to understand under what conditions inferences such as this one are allowed from first principles. More broadly, and motivated by the lack of a systematic treatment to combining partially-observed datasets, we formally introduce and study the problem of *causal effect identifiability under partial-observability*. More specifically, the main contributions of this work are as follows: (i) We develop novel machinery to account for partial-observability constraints, including constructs for co-identification, embedding of critical identification factors, and formal understanding of minimum viable embeddings. Putting these results together, we derive a novel graphical condition for identification under partial-observability; (ii) We then develop the first general algorithm that avoids redundant computations and runs in polynomial time for the known subclasses of identifiability problems under full-observability. Furthermore, we provide a detailed discussion on the necessity of our algorithm and the NP-completeness status of this particular identifiability problem. Finally, we discuss the extension of this work to the transportability setting (Bareinboim & Pearl, 2014) in which the corresponding datasets may come from multiple, heterogeneous domains.

One metaphor that will be informative and facilitate the understanding of this paper is to compare the task of identification under partial-observability to a jigsaw puzzle (Fig. 2). The targeted causal query and underlying causal graph define the puzzle and its layout (Sec. 4.1), while each of the available observational and experimental distributions provides pieces and chunks (pieces tied together) for the puzzle (Sec. 4.1.1 and 4.1.2). Then, the given pieces and chunks are combined, without overlapping each other, to complete the puzzle, eventually forming the targeted effect (Sec. 4.2). For simplicity, we start in Sec. 3 the discussion with a relatively simple class of puzzles, and then build over it, refining its understanding (just after the preliminaries).

2. Preliminaries

Following conventions in the field, a variable is denoted by an uppercase letter, e.g., Z, and its value is denoted by the corresponding lowercase letter, $z \in \mathfrak{X}_Z$, where $\mathfrak{X}_Z$ is the state space of Z. Bold letters are for a set of variables or values, e.g., $\mathbf{z} \in \mathfrak{X}_{\mathbf{Z}} = \times_{Z \in \mathbf{Z}} \mathfrak{X}_Z$. For simplicity, we may omit curly braces, e.g., $f(\{x\})$ versus $f(x)$, for a singleton set when it is used as an argument.

This paper builds on the language of Structural Causal Models (SCM) (Pearl, 2000). Each SCM $\mathcal{M}$ is a quadruple, $\mathbf{U}$, $\mathbf{V}$, $\mathbf{F}$, and $P(\mathbf{U})$. The set of unobserved variables $\mathbf{U}$ follows the joint probability distribution $P(\mathbf{U})$. The set of observed variables $\mathbf{V}$ are specified through the set of structural functions $\mathbf{F} = \{f_V\}_{V \in \mathbf{V}}$, where each function is of the form $f_V(\mathbf{pa}_V, \mathbf{u}_V)$ such that $\mathbf{pa}_V$ and $\mathbf{u}_V$ are the values for $\mathbf{PA}_V \subseteq \mathbf{V} \setminus \{V\}$ and $\mathbf{U}_V \subseteq \mathbf{U}$, respectively. The SCM $\mathcal{M}$ induces a causal graph $\mathcal{G}$ over $\mathbf{V}$ where there are directed edges $W \to V$ if $W \in \mathbf{PA}_V$ and bidirected ones $W \leftrightarrow V$ if there exists an unobserved confounder (UC, for short) $U \in \mathbf{U}_W \cap \mathbf{U}_V$. Further, $\mathcal{M}$ induces a set of observational and interventional distributions. One can intervene on $\mathbf{X}$, setting them to $\mathbf{x}$, which yields a submodel $\mathcal{M}_{\mathbf{x}}$, where the function for $X \in \mathbf{X}$ in $\mathcal{M}$ is replaced by constants $x \in \mathbf{x}$. The distribution generated by $\mathcal{M}_{\mathbf{x}}$ is denoted by $P_{\mathbf{x}}(\mathbf{V})$ (or $P(\mathbf{V} \mid do(\mathbf{x}))$). For simplicity, $P_{\mathbf{Z}}(\mathbf{W})$ denotes a collection of probabilities $\{P_{\mathbf{z}}(\mathbf{w})\}_{\mathbf{z} \in \mathfrak{X}_{\mathbf{Z}}, \mathbf{w} \in \mathfrak{X}_{\mathbf{W}}}$, which we may call $P_{\mathbf{Z}}(\mathbf{W})$ just a distribution. For a more detailed discussion on SCMs, please refer to (Pearl, 2000, Ch. 7).

We denote by $\mathbf{V}_{\mathcal{H}}$ the vertices of a graph $\mathcal{H}$. We often use $\mathbf{V}$ as the vertices of $\mathcal{G}$ where no ambiguity arises. We denote by $pa(\mathbf{W})_{\mathcal{G}}$ the union of parents of $W \in \mathbf{W}$ in $\mathcal{G}$. Similarly, ch, an, and de are children, ancestors, and descendants. Additionally, Ch, An, De include their arguments as well, e.g., $Ch(\mathbf{W})_{\mathcal{G}} = ch(\mathbf{W})_{\mathcal{G}} \cup \mathbf{W}$. A subgraph of $\mathcal{G}$ over $\mathbf{V}' \subseteq \mathbf{V}$ is denoted by $\mathcal{G}[\mathbf{V}']$. Further, $\mathcal{G}_{\overline{\mathbf{X}}}$ and $\mathcal{G}_{\underline{\mathbf{Z}}}$ denote $\mathcal{G}$ with edges incoming to $\mathbf{X}$ and going out from $\mathbf{Z}$ removed, respectively.

The *latent projection* (or projection, for short) of a causal

graph is defined in a way to retain the causal relationships among a subset of variables. This concept is crucial in understanding partial-observability. We define projection as below, adopted from (Tian & Pearl, 2003).

Definition 1 (Projection)**.** The projection of a causal graph $\mathcal{G}$ over $\mathbf{V}$ on $\mathbf{V}' \subseteq \mathbf{V}$, denoted by $\mathcal{G}\langle\mathbf{V}'\rangle$, is a causal graph over $\mathbf{V}'$ such that for every pair of vertices X and Y:

1. There exists a directed edge $X \to Y$ in $\mathcal{G}\langle\mathbf{V}'\rangle$ if there exists a directed path from X to Y in $\mathcal{G}$ such that every vertex other than X and Y on the path is not in $\mathbf{V}'$.
2. There exists a bidirected edge $X \leftrightarrow Y$ in $\mathcal{G}\langle\mathbf{V}'\rangle$ if there exists a divergent path[1] between X and Y in $\mathcal{G}$ such that every vertex other than X and Y on the path is not in $\mathbf{V}'$.

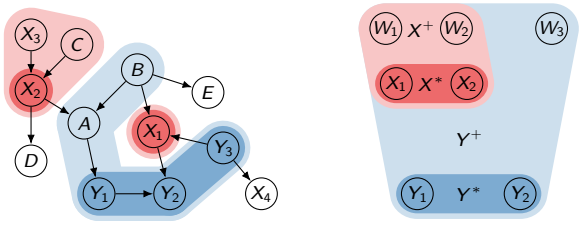
We denote by $\mathcal{G}\langle\text{-}\mathbf{W}\rangle$ a causal graph with $\mathbf{W}$ *projected out*, i.e., $\mathcal{G}\langle\text{-}\mathbf{W}\rangle = \mathcal{G}\langle\mathbf{V} \setminus \mathbf{W}\rangle$. We further write $\mathcal{G} \setminus \mathbf{W}$ for the *vertex-deleted* subgraph $\mathcal{G}[\mathbf{V} \setminus \mathbf{W}]$, in which the edges through $\mathbf{W}$ are *not* preserved; this is to be distinguished from the projection $\mathcal{G}\langle\text{-}\mathbf{W}\rangle$, which does preserve them. Conditional independence statements and the rules of do-calculus (Pearl, 1995) on a projection are valid in $\mathcal{G}$, vice versa. We often simplify notation involving projections by letting $\mathcal{G}' = \mathcal{G}\langle\mathbf{V}'\rangle$. Similarly, $\mathcal{G}''$ and $\mathcal{G}_j$ is defined for $\mathbf{V}''$ and $\mathbf{V}_j$, respectively. An illustration for how projection induces new edges and some remarks are provided in Sec. A.

Graphical Constructs for Identifiability Throughout the paper, we fix the use of a few symbols. A query $P_{\mathbf{x}}(\mathbf{y})$ is defined on a causal graph $\mathcal{G}$ of an unknown SCM $\mathcal{M}$ where its vertices (or variables) are $\mathbf{V}$ where $\mathbf{X}$ and $\mathbf{Y}$ are possibly empty disjoint subsets of $\mathbf{V}$. Further, we define a few new symbols that would greatly help understanding the anatomy of a given graph with respect to the identification of a query, namely:

$$\mathbf{X}^* = \mathbf{X} \cap An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}} \qquad \mathbf{X}^+ = An(\mathbf{X}^*)_{\mathcal{G}} \setminus \mathbf{Y}^+$$
$$\mathbf{Y}^* = \mathbf{Y} \qquad \mathbf{Y}^+ = An(\mathbf{Y})_{\mathcal{G}_{\underline{\mathbf{X}^*}}}$$
$$\mathbf{V}^* = \mathbf{X}^* \cup \mathbf{Y}^* \qquad \mathbf{V}^+ = \mathbf{X}^+ \cup \mathbf{Y}^+.$$

The left column presents essential parts of the query and the right column describes relevant variables in identifying the query; Fig. 3 provides an illustration of these notions. Their meanings and relationships will become more clear when they appear in lemmas and theorems, especially in Sec. 4. We will use the same color scheme to visualize graphs.

[1] A divergent path between X and Y exists if there are two vertex-disjoint directed paths $(W_1, \dots, X)$ and $(W_2, \dots, Y)$ in $\mathcal{G}$ towards X and Y, respectively, each possibly trivial (i.e., $W_1 = X$ or $W_2 = Y$), such that $W_1 \leftrightarrow W_2$; or if there is a vertex $W \notin \{X, Y\}$ together with two directed paths $(W, \dots, X)$ and $(W, \dots, Y)$ meeting only at W. Equivalently, a divergent path between X and Y is a path on which every vertex other than X and Y is a non-collider and which has an arrowhead at both X and Y.



(a) a causal graph (part) (b) abstract illustration

Figure 3. An illustrative example for $\mathbf{X}^+$ (light red), $\mathbf{Y}^+$ (light blue), $\mathbf{X}^*$ (dark red), and $\mathbf{Y}^*$ (dark blue). Bidirected edges are not relevant in defining those symbols and omitted on purpose.

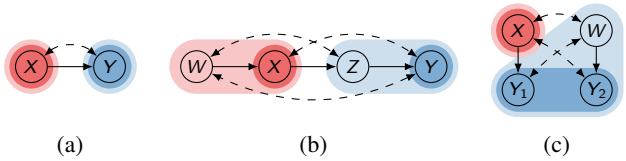
Concepts for Non-identifiability One of crucial building blocks for characterizing our problem of interest is a graphical structure articulating the non-identifiability of a causal effect. A graph $\mathcal{H}$ is said to be a *c-component* if a subset of its bidirected arcs forms a spanning tree over all vertices in $\mathcal{H}$ (Tian & Pearl, 2002; Tian, 2002). The definition can be understood as a set of vertices that are connected via bidirected edges. We utilize the notion of the decomposition of a set of vertices in a graph with respect to its maximal c-components, which we call c-component decomposition. In a special type of c-component, a graph $\mathcal{H}$ with root-set (sink nodes) $\mathbf{R}$ is said to be an $\mathbf{R}$-rooted *c-forest* if $\mathcal{H}$ is a c-component with a minimal number of edges.

Definition 2 (Hedge)**.** A hedge is a pair of $\mathbf{R}$-rooted c-forests $\langle\mathcal{F}, \mathcal{F}'\rangle$ such that $\mathcal{F}' \subseteq \mathcal{F}$.

A hedge is said to be formed for $P_{\mathbf{x}}(\mathbf{y})$ if $\mathbf{R} \subseteq An(\mathbf{Y})_{\mathcal{G}_{\underline{\mathbf{X}}}}$, $\mathcal{F} \cap \mathbf{X} \neq \emptyset$, and $\mathcal{F}' \cap \mathbf{X} = \emptyset$, which implies the non-identifiability of $P_{\mathbf{x}}(\mathbf{y})$ from $P(\mathbf{V})$ in $\mathcal{G}$ (Shpitser & Pearl, 2006a). We prefer to separate the graphical definition of hedge from the specific syntactic goal/task, following discussion in (Lee et al., 2019). Such hedge satisfies that $\mathbf{V}_{\mathcal{F}\setminus\mathcal{F}'} \subseteq \mathbf{V}^+$ intersects with $\mathbf{X}^*$. Further, $\mathbf{V}_{\mathcal{F}'} \subseteq \mathbf{Y}^+$ since $\mathbf{X}^+$ being the part of $\mathcal{F}'$ implies that $X \in \mathbf{X}^*$ is in $\mathcal{F}'$, which violates the definition. Examples are illustrated in Fig. 4 where the first two causal graphs are hedges for $P_x(y)$ but the last one isn't. Omitted proofs and other supporting materials are provided in the Appendix.

3. Identifiability with a Single Partially-Observed Distribution

We begin with a simpler, yet crucial identifiability problem that concerns with identifying a causal effect given a single partially-observed distribution. Using the jigsaw metaphor discussed earlier, this task can be seen as a puzzle where all



(a) (b) (c)

Figure 4. (a, b) Hedges for $P_x(y)$ where, for both cases, $\{Y\}$ is the sole root set and the only element of $\mathcal{F}'$ with $\mathcal{F} = \mathcal{G}$. (c) is not a hedge for $P_x(\mathbf{y})$ but $\{Y_1, W\}$-rooted c-forests $\mathcal{F} = \mathcal{G}[\{Y_1, W, X\}]$ and $\mathcal{F}' = \mathcal{G}[\{Y_1, W\}]$ form a hedge for $P_x(\mathbf{y})$.

the available chunks are of the same color, as defined next.

Definition 3 (Causal Effect Identifiability under a Partially-Observed Distribution)**.** Given a causal graph $\mathcal{G}$, let $\mathbf{Z}'$, $\mathbf{V}', \mathbf{X}, \mathbf{Y} \subseteq \mathbf{V}$ where $\mathbf{Z}' \cap \mathbf{V}' = \emptyset$ and $\mathbf{X} \cap \mathbf{Y} = \emptyset$. The causal effect $P_{\mathbf{x}}(\mathbf{y})$ is said to be identifiable from $P_{\mathbf{Z}'}(\mathbf{V}')$ in $\mathcal{G}$ if $P_{\mathbf{x}}(\mathbf{y})$ is (uniquely) computable from $P_{\mathbf{Z}'}(\mathbf{V}')$ in any causal model which induces $\mathcal{G}$ and $P_{\mathbf{Z}'}(\mathbf{V} \setminus \mathbf{Z}') > 0$.

The definition is similar to a number of identifiability problems with full-observability except for the given data being partially-observed. The positivity assumption is imposed on $P_{\mathbf{Z}'}(\mathbf{V} \setminus \mathbf{Z}')$ (i.e., the full joint before the projection) instead of the given distribution $P_{\mathbf{Z}'}(\mathbf{V}')$. This is to ensure that the partial-observability is correctly responsible for the non-identifiability of a causal effect especially when the effect is identifiable with $P_{\mathbf{Z}'}(\mathbf{V} \setminus \mathbf{Z}')$.

We show multiple necessary criteria as follows.

Proposition 1. *A causal effect $P_{\mathbf{x}}(\mathbf{y})$ is identifiable from $P_{\mathbf{Z}'}(\mathbf{V}')$ in $\mathcal{G}$ only if*

- $\mathbf{Y} \subseteq \mathbf{V}'$ *(inclusion of outcomes),*
- $\mathbf{X}^* \subseteq \mathbf{V}' \cup \mathbf{Z}'$ *(inclusion of minimal treatments), and*
- $\mathbf{Z}' \cap \mathbf{Y}^+ = \emptyset$ *(undisturbed mechanisms).*

Under full-observability, the first condition is implied by the third condition, and the second condition holds trivially since $\mathbf{Z}' \cup \mathbf{V}' = \mathbf{V}$ and $\mathbf{X}, \mathbf{Y} \subseteq \mathbf{V}$. The third condition highlights that an intervention prohibits the understanding of the underlying natural mechanisms relevant to the distribution over $\mathbf{Y}$ (Lee et al., 2019). When these criteria hold, the problem is reducible to the classic identifiability.

Lemma 1. *Given $P_{\mathbf{x}}(\mathbf{y})$ and $P_{\mathbf{Z}'}(\mathbf{V}')$ in $\mathcal{G}$ satisfying the three criteria in Prop. 1, $P_{\mathbf{x}}(\mathbf{y})$ is identifiable from $P_{\mathbf{Z}'}(\mathbf{V}')$ in $\mathcal{G}$ if and only if $Q_{\mathbf{x}^* \setminus \mathbf{Z}'}(\mathbf{y})$ is identifiable in $(\mathcal{G} \setminus \mathbf{Z}')\langle \mathbf{V}' \cap \mathbf{V}^+ \rangle$ where $Q = P_{\mathbf{z}'}$ for any $\mathbf{z}'$ that agrees with $\mathbf{x}$ on $\mathbf{X}^* \cap \mathbf{Z}'$.*

Roughly speaking, Lemma 1 implies that the effect is identifiable when there is no structure that embeds a hedge under the projection onto the partially-observed variables. A simple example is provided (Fig. 5) showing the identifiability of $P_x(y)$ given $P_Z(X, Y)$. The effect is not identifiable with $P(\mathbf{V})$ (the existence of a hedge with $\mathcal{F} = \mathcal{G}$, $\mathcal{F}' = \mathcal{G}[\{W, Y\}]$), $P_W(X, Y, Z)$ (a disturbed mechanism), nor $P_Y(W, X, Z)$ (exclusion of outcomes).

Corollary 1 (Soundness and Completeness)**.** *$P_{\mathbf{x}}(\mathbf{y})$ is identifiable from $P_{\mathbf{Z}'}(\mathbf{V}')$ in $\mathcal{G}$ if and only if $\mathbf{Y} \subseteq \mathbf{V}'$, $\mathbf{X}^* \subseteq \mathbf{V}' \cup \mathbf{Z}'$, $\mathbf{Z}' \cap \mathbf{Y}^+ = \emptyset$, and $Q_{\mathbf{x}^* \setminus \mathbf{Z}'}(\mathbf{y})$ is identifiable in $(\mathcal{G} \setminus \mathbf{Z}')\langle \mathbf{V}' \cap \mathbf{V}^+ \rangle$ where $Q = P_{\mathbf{z}'}$ for any $\mathbf{z}'$ that agrees with $\mathbf{x}$ on $\mathbf{X}^* \cap \mathbf{Z}'$.*

4. Identifiability with Multiple Partially-Observed Distributions

We now investigate the main task of this paper, i.e., how to systematically use multiple distributions with different levels of observability, as defined next.

Definition 4 (Causal Effect Identifiability under Partially-Observed Distributions (GID-PO))**.** Let $\mathcal{G}$ be a causal graph, and $\mathbf{X}, \mathbf{Y} \subseteq \mathbf{V}$ be sets corresponding to the treatment and outcomes variables, respectively. Further, let $\mathbb{P} = \{P_{\mathbf{Z}_i}(\mathbf{V}_i)\}_{i=1}^m$ be a collection of partially observable distributions such that $\mathbf{Z}_i$ and $\mathbf{V}_i$ are disjoint subsets of $\mathbf{V}$ for $1 \leq i \leq m$. The causal effect $P_{\mathbf{x}}(\mathbf{y})$ is identifiable from the graph $\mathcal{G}$ and $\mathbb{P}$ if it is uniquely computable from $\mathbb{P}$ in any model that induces $\mathcal{G}$ where $\{P_{\mathbf{Z}_i}(\mathbf{V} \setminus \mathbf{Z}_i)\}_{i=1}^m$ are positive distributions.

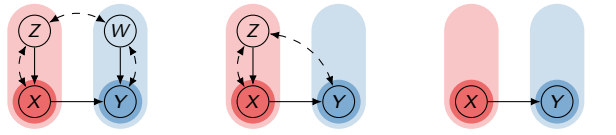
This definition generalizes g-identifiability (GID, Lee et al., 2019) with partial-observability. An example of the problem is shown in Fig. 7a, where $P(B, C, Y)$ and $P_X(A, C)$ are given to identify $P_x(y)$ (the same as Fig. 1c). Its formula can be derived as (see Sec. D for the detailed derivation):

$$P_x(y) = \sum_{a,c} P_{x'}(a) P_x(c|a) \sum_b P(y|b, c) P(b), \quad (1)$$

where x' can be any value in $\mathfrak{X}_X$. Naively incorporating Cor. 1 into GID, which looks for colored pieces but not chunks, fails to identify the query since one of the pieces is not identifiable with any of the given datasets (Fig. 7c).

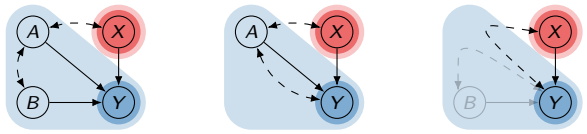
Lemma 2. *Let $\mathcal{G}$ be a causal graph and $\mathbf{X}, \mathbf{Y} \subseteq \mathbf{V}$. A query $P_{\mathbf{x}}(\mathbf{y})$ is not identifiable if there exist two causal models $\mathcal{M}^1$ and $\mathcal{M}^2$ compatible with $\mathcal{G}$ such that $P^1_{\mathbf{Z}_i}(\mathbf{V}_i) = P^2_{\mathbf{Z}_i}(\mathbf{V}_i)$, for every $P_i \in \mathbb{P}$, but $P^1_{\mathbf{x}}(\mathbf{y}) \neq P^2_{\mathbf{x}}(\mathbf{y})$, $P^1_{\mathbf{Z}_i}(\mathbf{V} \setminus \mathbf{Z}_i) > 0$, and $P^2_{\mathbf{Z}_i}(\mathbf{V} \setminus \mathbf{Z}_i) > 0$.*

Consider a causal graph (Fig. 6a) where two observational distributions $P(X, Y, A)$ and $P(B, X, Y)$ are available. A causal query $P_x(y)$ is identifiable given $P(X, Y, A, B)$ but not from *each* of $P(X, Y, A)$ and $P(B, X, Y)$ due to the existence of a hedge (Cor. 1). Further, one can show that the query is not identifiable taking *both* into account: Let $f_X = U_1$, $f_A = U_1 \oplus U_2 \oplus U_A$, $f_B = U_2$ be the common functions between $\mathcal{M}^1$ and $\mathcal{M}^2$, and $f^1_Y = X \oplus A \oplus B \oplus U_Y$



(a) $\mathcal{G}$ (b) $\mathcal{G}\langle\text{-}W\rangle$ (c) $(\mathcal{G}\setminus\{Z\})\langle\text{-}W\rangle$

Figure 5. Causal diagrams representing the identifiability of $P_x(y)$ given $P_Z(X,Y)$ as $P_x(y)=P_z(y|x)$ for any z.



(a) $\mathcal{G}$ (b) $\mathcal{G}\langle\text{-}B\rangle$ (c) $\mathcal{G}\langle\text{-}A\rangle$

Figure 6. A causal graph $\mathcal{G}$ (a) which embeds hedges under different projections onto (b) $\{A,X,Y\}$ and (c) $\{B,X,Y\}$.

and $f_Y^2=U_Y$. Further, let U_1 and U_2 be two fair coins, let $P(U_A=1)=\epsilon$ and $P^1(U_Y=1)=\epsilon$ with $0<\epsilon<0.5$, and let $P^2(U_Y=1)=2\epsilon(1-\epsilon)$. Then, $\mathcal{M}^1$ and $\mathcal{M}^2$ will agree on both $P(A,X,Y)$ and $P(B,X,Y)$, while $P^1_{X=0}(Y=0)=0.5$ and $P^2_{X=0}(Y=0)=1-2\epsilon(1-\epsilon)$; at $\epsilon=0.1$ these are 0.5 and 0.82. The private noise U_A is what makes the joint $P(A,B,X,Y)$ strictly positive as Lemma 2 requires: without it $A=X\oplus B$ holds deterministically and half of the cells vanish.

4.1. Characterization of Factors under Projection

To develop an algorithm capable of combining the different parts of the available distributions, we briefly review approaches to the related problem of decomposing distributions currently known in the literature. A joint probability distribution $P(\mathbf{V})$ in $\mathcal{G}$ can be seen as the product of causal effects in $\mathcal{G}$. Tian & Pearl (2002) introduced the *Q-decomposition* that expresses $P(\mathbf{v})$ using *c-factors* as follows: $P(\mathbf{v})=\prod_i Q_i=\prod_i P_{\mathbf{v}\setminus\mathbf{s}_i}(\mathbf{s}_i)$, where $\mathbf{S}_i$ is a c-component of $\mathbf{V}$.[2] We will characterize here such factorization under projections. As a first step, we define the key notion of generalized c-factors:

Definition 5 (gc-factors)**.** Let $\mathbf{V}'$ be a subset of variables such that $\mathbf{V}^*\subseteq\mathbf{V}'\subseteq\mathbf{V}^+$ and let $\mathcal{G}'=\mathcal{G}\langle\mathbf{V}'\rangle$. Let $\mathbb{S}=\{\mathbf{S}_i\}_{i=1}^k$ be the c-components of $\mathcal{G}'[\mathbf{Y}^{+\prime}]$ where $\mathbf{Y}^{+\prime}=An(\mathbf{Y})_{\mathcal{G}'_{\underline{\mathbf{X}^*}}}$. Then, the *gc-factors* of $P_{\mathbf{x}}(\mathbf{y})$ in $\mathcal{G}$ with respect to $\mathbf{V}'$ is defined as

$$\mathbb{F}^{\mathbf{X},\mathbf{Y}}_{\mathcal{G},\mathbf{V}'}=\{\langle pa(\mathbf{S}_i)_{\mathcal{G}'}\setminus\mathbf{S}_i,\mathbf{S}_i\rangle\}_{i=1}^k.$$

For example, consider a causal graph $\mathcal{G}$ in Fig. 6a and a query $P_x(y)$, where $\mathbf{V}^*=\{X,Y\}$ and $\mathbf{V}^+=\mathbf{V}$. With $\mathbf{V}'=\mathbf{V}$, $\mathbf{Y}^{+\prime}=\{A,B,Y\}$ (i.e., variables in blue), and $\mathbb{S}=\{\{A,B\},\{Y\}\}$ (i.e., the two c-components in the blue area). Hence, the gc-factors $\mathbb{F}^{X,Y}_{\mathcal{G},\mathbf{V}}=\{\langle\emptyset,\{A,B\}\rangle,\langle\{A,B,X\},\{Y\}\rangle\}$. With $\mathbf{V}'=\{A,X,Y\}$, $\mathcal{G}'$ is shown in Fig. 6b. In this case, $\mathbf{Y}^{+\prime}=\{A,Y\}$, $\mathbb{S}=\{\{A,Y\}\}$, and $\mathbb{F}^{X,Y}_{\mathcal{G},\{A,X,Y\}}=\{\langle\{X\},\{A,Y\}\rangle\}$.

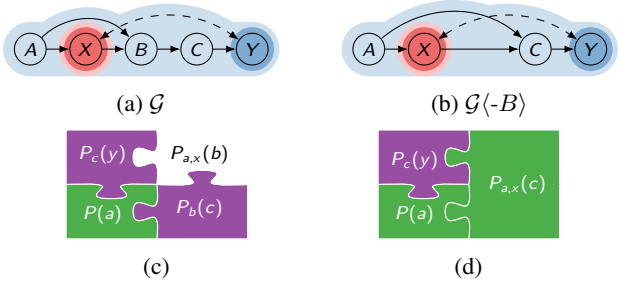
A gc-factor (or, simply, a factor), which is represented as a pair of sets of variables, can be used to present a set of distributions, e.g., $\langle\mathbf{X}_i,\mathbf{Y}_i\rangle\in\mathbb{F}^{\mathbf{X},\mathbf{Y}}_{\mathcal{G},\mathbf{V}}$ and $P_{\mathbf{X}_i}(\mathbf{Y}_i)$. We may call the first and second elements as $\mathbf{X}$- and $\mathbf{Y}$-side of the gc-factor, respectively. Graphically, this decomposes $\mathbf{Y}^{+\prime}$ in $\mathcal{G}'$ with respect to c-components so that the $\mathbf{Y}$-side of every gc-factor is maximally confounded with respect to the underlying projection. The $\mathbf{X}$-side resides in $\mathbf{Y}^{+\prime}\cup\mathbf{X}^*$.

For brevity, we simplify the notation by $\mathbb{F}=\mathbb{F}^{\mathbf{X},\mathbf{Y}}_{\mathcal{G},\mathbf{V}^+}$ — the finest level admitted by the definition, since gc-factors are defined only for $\mathbf{V}^*\subseteq\mathbf{V}'\subseteq\mathbf{V}^+$ — and a superscript is delegated to $\mathbf{V}$ so that $\mathbb{F}'=\mathbb{F}^{\mathbf{X},\mathbf{Y}}_{\mathcal{G},\mathbf{V}'}$. Also we interchangeably use a factor $\langle\mathbf{Z},\mathbf{W}\rangle$ with $P_{\mathbf{Z}}(\mathbf{W})$ or with $P_{\mathbf{z}}(\mathbf{w})$ for an arbitrary assignment. Probabilities associated with the gc-factors of $P_{\mathbf{x}}(\mathbf{y})$ in $\mathcal{G}$ with respect to $\mathbf{V}'$ can form an expression for $P_{\mathbf{x}}(\mathbf{y})$.

Proposition 2 (gc-decomposition)**.** *For* $\mathbf{V}^*\subseteq\mathbf{V}'\subseteq\mathbf{V}^+$,

$$P_{\mathbf{x}}(\mathbf{y})=\textstyle\sum_{\mathbf{y}^{+\prime}\setminus\mathbf{Y}}\prod_{\{i|\langle\mathbf{X}_i,\mathbf{Y}_i\rangle\in\mathbb{F}'\}}P_{\mathbf{x}_i}(\mathbf{y}_i)\qquad(2)$$

The algorithm for general-identifiability (Lee et al., 2019) makes the use of this decomposition based on $\mathbb{F}$ (i.e., restricted to $\mathbf{V}'=\mathbf{V}^+$) and identifies each factor using one of the available distributions. However, such strategy relying



(a) $\mathcal{G}$ (b) $\mathcal{G}\langle\text{-}B\rangle$ (c) (d)

Figure 7. Causal graphs (a) $\mathcal{G}$ and (b) $\mathcal{G}\langle\text{-}B\rangle$. Corresponding gc-factors (c, d) represented as puzzle where identified factors are colored with green, $P_X(A,C)$, and purple, $P(B,C,Y)$.[4]

[2] Such a factorization leads to a natural divide-and-conquer approach to the problem of causal identification, which underpins, sometimes more or less explicitly, most of the results in this literature (e.g., (Huang & Valtorta, 2006a; Shpitser & Pearl, 2006a; Bareinboim & Pearl, 2012a; 2014; Lee et al., 2019), to cite a few). Depending on the identification task, one can prove that the Q-decomposition leads to a complete characterization.

[4] In general, 2-dimensional illustrations cannot faithfully capture every aspect of causal jigsaw puzzles.

on the decomposition based on $\mathbb{F}$ is insufficient for handling partially-observed distributions. Recall Fig. 7a where $P_x(y)$ can be factorized as

$$P_x(y) = \sum_{a,b,c} P(a)P_{a,x}(b)P_b(c)P_c(y), \tag{3}$$

following Prop. 2. Unfortunately, $P_{a,x}(b) = P(b|a,x)$ is not identifiable from each of available distributions since no distribution includes all A, B, X (Cor. 1). On the other hand, a gc-decomposition based on a projection onto $\mathbf{V}' = \{A, C, X, Y\}$ (Fig. 7b) yields

$$P_x(y) = \sum_{a,c} P(a)P_{a,x}(c)P_c(y), \tag{4}$$

which allows each factor to be identified by at least one of the available distributions (i.e., $P(a) = P_{x'}(a)$, for any $x' \in \mathfrak{X}_X$, $P_{a,x}(c) = P_x(c|a)$, and $P_c(y) = \sum_b P(y|b,c)P(b)$). This provides a basis to the following sufficient condition.

Lemma 3 (Soundness)**.** *Let $\mathcal{G}$, $P_{\mathbf{x}}(\mathbf{y})$, $\mathbb{P}$ be the causal graph, the query, and the distributions forming a GID-PO instance. If there exists a subset of variables $\mathbf{V}^* \subseteq \mathbf{V}' \subseteq \mathbf{V}^+$ such that every term $P_{\mathbf{x}_i}(\mathbf{y}_i)$ in each gc-factor $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$ in $\mathbb{F}'$ is identifiable from $P \in \mathbb{P}$, then $P_{\mathbf{x}}(\mathbf{y})$ is identifiable from $\mathbb{P}$ in $\mathcal{G}$.*

Given Lemma 3, we are interested in finding $\mathbf{V}'$, a subset of $\mathbf{V}$, which would yield a gc-decomposition where each gc-factor is identifiable.[5] We note that a natural solution emerges since one could search over an exponential number of subsets of $\mathbf{V}^+$. While this would certainly lead to a sound procedure, it is clearly the case that this offers little to no insight into the problem. This motivates us to study the relationships among gc-factors at different levels of projections so as to develop an efficient solution while avoiding this naive, and clearly intractable solution.

4.1.1. Embedding Factor and Co-identification

In this section, we formally relate gc-factors before and after marginalizations through two new concepts called *embedding factor* and *co-identification*, which associate factors and their identification under various levels of projections so as for us to perceive the task of identification from a more comprehensive view.

Comparing the decompositions in Eqs. (3) and (4) based on $\mathcal{G}$ and $\mathcal{G}\langle\text{-}B\rangle$, respectively, we observe that two factors $P(a)$ and $P_c(y)$, which does not have B in it, are shared while other two factors $P_{a,x}(b)$ and $P_b(c)$ in Eq. (3) are replaced to $P_{a,x}(c)$ in Eq. (4) where we can elicit $\sum_b P_{a,x}(b)P_b(c) = P_{a,x}(c)$. We characterize such changes in gc-factors before and after a projection (equivalently, a marginalization).

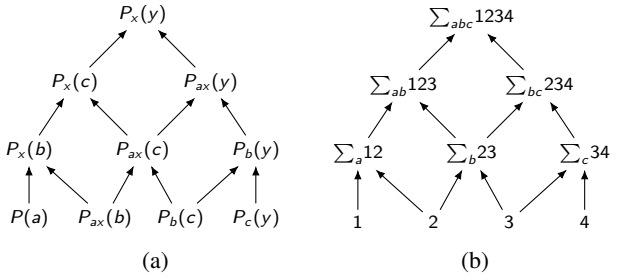


Figure 8. (a) embedding relationships among gc-factors where a directed edge $i \to j$ indicates that j embeds i (represented with a transitive reduction). (b) the same embedding relationships with a sum-product notation.

Proposition 3 (Factors under Marginalization)**.** *Let $\mathbf{W} \subset \mathbf{V}'$ and $\mathbf{V}'' = \mathbf{V}' \setminus \mathbf{W}$ where $\mathbf{W} \cap \mathbf{V}^* = \emptyset$. Let $\mathcal{H}$ be an undirected graph where vertices are $\langle \mathbf{X}'_j, \mathbf{Y}'_j \rangle \in \mathbb{F}'$ and an edge exists if two factors satisfy their $\mathbf{Y}$-sides intersecting with $Ch(W)_{\mathcal{G}'}$ for some $W \in \mathbf{W}$. Then, vertices (i.e., gc-factors) in each connected component of $\mathcal{H}$ are merged to form $\langle \mathbf{X}''_k, \mathbf{Y}''_k \rangle \in \mathbb{F}''$ such that*

$$\mathbf{Y}''_k = \left(\cup_{j \in \mathbf{j}} \mathbf{Y}'_j\right) \setminus \mathbf{W}, \quad \mathbf{X}''_k = \left(\cup_{j \in \mathbf{j}} \mathbf{X}'_j\right) \setminus \mathbf{Y}''_k \setminus \mathbf{W},$$

where $\mathbf{j}$ is the set of indices of gc-factors in $\mathbb{F}'$ forming the connected component. Other gc-factors without $\mathbf{W}$ are remained intact, shared by both $\mathbb{F}'$ and $\mathbb{F}''$.

For instance, marginalizing B out in Eq. (3) will consolidate factors whose $\mathbf{Y}$-sides intersecting with $Ch(B)_{\mathcal{G}}=\{B, C\}$, that is, $P_{a,x}(b)$ and $P_b(c)$. Further, $\sum_b P_{a,x}(b)P_b(c)$ will result in a new gc-factor that has its $\mathbf{Y}$-side, $(\{B\}\cup\{C\})\setminus\{B\}=\{C\}$ and $\mathbf{X}$-side, $(\{A, X\}\cup\{B\})\setminus\{C\}\setminus\{B\}=\{A, X\}$. We introduce an embedding relationship to describe the connection between the factors merged through a projection and a resulting factor.

Definition 6 (Embedding Factor)**.** A gc-factor $\langle \mathbf{X}', \mathbf{Y}' \rangle \in \mathbb{F}'$ is said to be *embedded in* a gc-factor $\langle \mathbf{X}'', \mathbf{Y}'' \rangle \in \mathbb{F}''$ for $\mathbf{V}^* \subseteq \mathbf{V}'' \subseteq \mathbf{V}'$ if $\mathbf{Y}''$ contains any variable in the $\mathbf{Y}$ parts of gc-factors in the connected component containing $\langle \mathbf{X}', \mathbf{Y}' \rangle$ in $\mathcal{H}$, which is constructed as follows. $\mathcal{H}$ is an undirected graph of gc-factors in $\mathbb{F}'$ where there exists an edge between two factors, say $\langle \mathbf{X}'_i, \mathbf{Y}'_i \rangle$, $\langle \mathbf{X}'_j, \mathbf{Y}'_j \rangle$, if both share $\mathbf{V}' \setminus \mathbf{V}''$, that is,

$$(\mathbf{X}'_i \cup \mathbf{Y}'_i) \cap (\mathbf{X}'_j \cup \mathbf{Y}'_j) \cap (\mathbf{V}' \setminus \mathbf{V}'') \neq \emptyset.$$

Note that a gc-factor is also a (non-proper) embedding factor of itself, and, whenever $\mathbf{Y}$ is a single c-component of $\mathcal{G}\langle\mathbf{V}^*\rangle$, the coarsest gc-factor $\langle \mathbf{X}^*, \mathbf{Y} \rangle$ (i.e., $P_{\mathbf{x}^*}(\mathbf{y}) = P_{\mathbf{x}}(\mathbf{y})$) is an embedding factor of every gc-factor. We illustrate embedding relationships among every gc-factor of

[5] After submitting this work (in early February, 2020), Lee & Shpitser (2020) independently introduced the problem of 'mID', where 'm' stands for marginal distributions, which corresponds to the notion of partial-observability here (Def. 4). Even though there are subtle differences in terminology and notation, their main result in this context (Lemma 3) can be seen as our Lemma 3.

$P_x(y)$ in $\mathcal{G}$ (Fig. 7a) in Fig. 8a. The four factors at the bottom correspond to $\mathbb{F}$ and the corresponding embedding relationships are represented as directed edges. For instance, $\langle\{A, X\}, \{C\}\rangle$ (i.e., $P_{a,x}(c)$) is an embedding factor of $\langle\{B\}, \{C\}\rangle$ and $\langle\{A, X\}, \{B\}\rangle$ (Fig. 7a). Quantitatively, embedding relationships can be understood as sum-product relationships (Fig. 8b), where projections are equivalent to marginalizations.

Although a gc-factor in $\mathbb{F}$ (a finest-grained factor) cannot be identified, it can be *identified as a group* under the summation, i.e., its embedding factor (a coarser-grained factor) is identified.

Definition 7 (Co-Identification)**.** If a factor Q is identifiable with a distribution, the factors in $\mathbb{F}$ embedded in Q are said to be *co-identified* with the distribution with respect to Q.

With the puzzle metaphor, co-identification of a factor refers to whether there will be a colored chunk which covers the piece. Following the strategy described in Lemma 3, $P_{\mathbf{x}}(\mathbf{y})$ will only be identified when every gc-factor in $\mathbb{F}$ is co-identified by one of the available distributions. However, there can be an exponential number of embedding factors for a gc-factor in $\mathbb{F}$. Hence, we investigate the relationship between (non-)identifiability of embedding factors.

4.1.2. Minimum Viable Embedding Factor

We introduce the crucial concept of *minimum viable embedding factor* (MVEF) that will help with the characterization of the co-identification of gc-factors in $\mathbb{F}$ with respect to a single available distribution. Its purpose is to find the finest-grained factors that a (partially-observed) distribution might be able to identify. In other words, as the default decomposition offers the factors ($\mathbb{F}$) of the right granularity for fully-observed distributions, we attempt to find factors of appropriate granularity with respect to a single partially-observed distribution.

Definition 8 (Minimum Viable Embedding Factor (MVEF))**.** An embedding factor $\langle\mathbf{X}^\dagger, \mathbf{Y}^\dagger\rangle \in \mathbb{F}^\dagger$ of a gc-factor $\langle\mathbf{X}_i, \mathbf{Y}_i\rangle \in \mathbb{F}$ is said to be a *minimum viable embedding factor* of $\langle\mathbf{X}_i, \mathbf{Y}_i\rangle$ with respect to $P_{\mathbf{Z}'}(\mathbf{V}')$ if the three criteria holds true: $\mathbf{Y}^\dagger \subseteq \mathbf{V}'$; $\mathbf{X}^\dagger \subseteq \mathbf{V}' \cup \mathbf{Z}'$; and $\mathbf{Z}' \cap An(\mathbf{Y}^\dagger)_{\mathcal{G}_{\underline{\mathbf{X}^\dagger}}} = \emptyset$, and $\mathbf{V}^+ \setminus \mathbf{V}^\dagger$ is minimal. Further, $\mathbf{V}^+ \setminus \mathbf{V}^\dagger$ is said to be a *MVEF-admissible set*.

Given one of available distributions $\mathbb{P}$, we want to check whether it co-identifies a gc-factor in $\mathbb{F}$. Among an exponential number of its embedding factors, we can choose an embedding factor, which satisfies the necessary conditions as specified in Prop. 1. A polynomial time algorithm for finding out an MVEF is depicted in Sec. D. The algorithm iteratively seeks variables to be projected out in order to satisfy the necessary conditions.

For example, take a look at a graph Fig. 9a where there are four factors in $\mathbb{F}$ (Fig. 9d). Consider co-identifying a factor $P_d(e, y_1)$ (with its $\mathbf{Y}$ side highlighted in Fig. 9a) with $P_B(A, X, \mathbf{Y}) \in \mathbb{P}$. Since $D \notin \mathbf{V}' \cup \mathbf{Z}' = \{B, A, X, \mathbf{Y}\}$ and $E \notin \mathbf{V}' = \{A, X, \mathbf{Y}\}$ — D lies on the treatment side of the factor and E on its outcome side — (Prop. 1), we may project out both D and E in $\mathcal{G}$ at once, and examine the resulting embedding factor of $P_d(e, y_1)$ in $\mathcal{G}\langle\text{-}\{E, D\}\rangle$ (Fig. 9c), that is, $P_{a,b,x}(\mathbf{y})$. The puzzle diagram illustrates how projecting out D and E, or equivalently, $\sum_{d,e}$ yields a chunk with three pieces except $P(a, b)$. Further, a Hasse diagram (Fig. 9e) represents the transitive reduction of embedding relationships where one can examine that $P_{a,b,x}(\mathbf{y})$ embeds the three factors in $\mathbb{F}$ excluding $P(a, b)$ and other two intermediate factors. Another example is given in Appendix showing that obtaining an MVEF may take multiple steps.

An MVEF, if exists, is uniquely determined. Further, MVEFs obtained given a distribution for a subset of gc-factors are disjoint with respect to embedded gc-factors.

Proposition 4 (Uniqueness)**.** *If an MVEF exists for a gc-factor with respect to a distribution, then it is unique.*

Proposition 5 (Disjointness)**.** *Given a distribution, let Q and R be MVEFs of two gc-factors in $\mathbb{F}$, respectively. Then either $Q = R$ or there is no common gc-factor embedded in Q and R.*

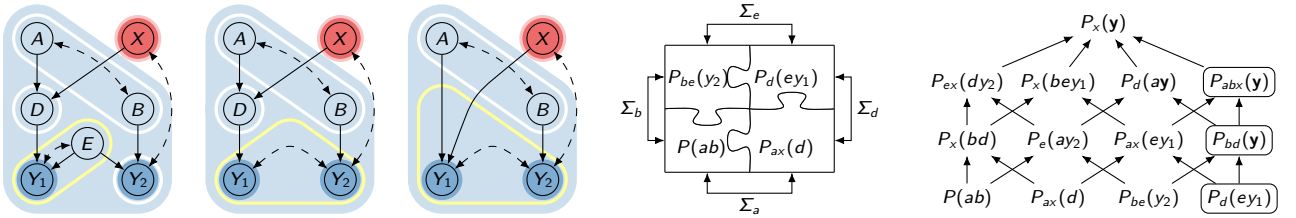
MVEFs are only a subset of all factors at different levels. If they are identifiable with a distribution, then they correspond to same-colored chunks. Should we further examine the possibility of other chunks of the same color? If an MVEF is found and not identified, is it still possible for factors further embedding the MVEF to be identifiable? We show the non-identifiability of factors embedding a non-identified MVEF.

Lemma 4 (Hedge over Embedding Factors)**.** *Consider a gc-factor $\langle\mathbf{X}_i, \mathbf{Y}_i\rangle \in \mathbb{F}$ and a distribution $P_{\mathbf{Z}'}(\mathbf{V}')$. The gc-factor is not co-identifiable with $P_{\mathbf{Z}'}(\mathbf{V}')$ with respect to any of its embedding factors if its MVEF does not exist or there exists a hedge for $P_{\mathbf{x}^\dagger \setminus \mathbf{Z}'}(\mathbf{y}^\dagger)$ in $(\mathcal{G} \setminus \mathbf{Z}')\langle\mathbf{V}'\rangle$ given the MVEF $\langle\mathbf{X}^\dagger, \mathbf{Y}^\dagger\rangle \in \mathbb{F}^\dagger$.*

Hence, the failure to identify an MVEF informs us that none of its embedding factors needs to be examined for (non-)identifiability. Below is a complementing lemma that, with puzzle terms, any chunks of the same color is composed of MVEFs (i.e., the smallest chunks of the color).

Lemma 5 (Compositionality)**.** *Given a distribution $P_{\mathbf{Z}'}(\mathbf{V}')$, any identifiable embedding factor of a gc-factor in $\mathbb{F}$ can be represented as the summation over the product of a subset of identified MVEFs.*

For instance, given that $P(a)$ and $P_{a,x}(c)$ are identified



(a) $\mathcal{G}$ (b) $\mathcal{G}\langle\text{-}E\rangle$ (c) $\mathcal{G}\langle\text{-}\{E, D\}\rangle$ (d) $\mathbb{F}$ (e) Hasse diagram

Figure 9. (a, b, c) Illustrations of causal diagrams with c-components in $\mathbf{Y}^{+\prime}$ highlighted which correspond to $\mathbf{Y}$-side of gc-factors, (d) $\mathbb{F}$ as a puzzle and how pieces are merged by marginalization, (e) a Hasse diagram for the embedding relationships among factors (not exhaustive) where four vertices at the bottom are $\mathbb{F}$. The *yellow-bordered* areas in (a, b, c) correspond to the right three vertices in (e); the remaining highlighted areas are the other c-components.

with $P_X(A, C)$ (Fig. 7d), it is clear to see the identifiability of $P_x(c)$ as $\sum_a P(a)P_{a,x}(c)$ and checking it is redundant. Thus, finding MVEFs and checking their identifiability are sufficient to comprehend how one distribution contributes to co-identification of a portion of factors.

4.2. Algorithm for GID-PO

Previous sections discussed what factors of different levels of projections will be available from each distribution. Hence, identified MVEFs from available distributions become chunks of different colors, which needs to be put together to complete the causal jigsaw puzzle $P_{\mathbf{x}}(\mathbf{y})$.

We devise a two-phase algorithm called GID-PO (Alg. 1), which first identifies MVEFs co-identifying gc-factors in $\mathbb{F}$ (colorful chunks), and then combine them to produce a formula for $P_{\mathbf{x}}(\mathbf{y})$. The implementation of the first phase is straightforward given the characterizations of MVEFs in the previous section. For every gc-factor in $\mathbb{F}$ and a distribution in $\mathbb{P}$ (Line 3), the existence of MVEF is first checked (Line 5); Lines 4 and 6 merely de-duplicate, skipping a pair whose MVEF has already been recorded from the same data (Line 4) or from any data (Line 6), and never discard a candidate. If it exists and identifiable, then record information including the MVEF and the gc-factors embedded in the MVEF (i.e., co-identified), the used distribution, and what variables are marginalized out (Lines 7–9).

With the colorful chunks of identified MVEFs, the second phase of the algorithm tries to complete the big picture. Based on Prop. 2, it attempts to select MVEFs whose co-identified gc-factors (represented as their indices in $\mathbb{F}$) do not overlap. This reduces to solving an *exact cover* (Line 13), an NP-complete problem (Karp, 1972): given a family $\mathbb{I}$ of subsets of a set $[n] = \{1, \dots, n\}$, whether there exists a subfamily $\mathbb{I}' \subseteq \mathbb{I}$ such that sets in $\mathbb{I}'$ are disjoint and $\cup\mathbb{I}' = \cup\mathbb{I} = [n]$. If a solution exists, then an expression for $P_{\mathbf{x}}(\mathbf{y})$ is written as the summation of product of sub-formulas corresponding to selected MVEFs where the variables to be marginalized follows Eq. (2) except that the part of variables are moved inside the sub-formulas (Line 16). Finally, we show next that this approach is indeed correct.

Theorem 1 (Soundness)**.** GID-PO *is sound.*

Remarkably, GID-PO is more efficient than a naive implementation of Lemma 3, which would require to run a traditional identifiability algorithm ($O(|\mathbf{V}|^4)$), for exponentially many gc-factors and for each dataset. On the other hand, GID-PO requires for the identifiability algorithm to run $|\mathbb{F}|$ times for each distribution to collect the identified MVEFs. Then, an exact cover over the universe $\{1, \dots, |\mathbb{F}|\}$, with the identified MVEFs as candidate sets, runs in time $O(2^{\ell}|\mathbb{J}|)$ by dynamic programming over subsets, i.e., exponential only in $|\mathbb{F}|$. (The number of distinct identified MVEFs is at most ℓm, and the number of distinct index sets of embedded gc-factors that they represent is at most $\min(\ell m, 2^{\ell} - 1)$; neither is bounded by $|\mathbb{F}|$.) More specifically, GID-PO will run in $O(\ell m|\mathbf{V}|^4 + \ell m 2^{\ell})$ while a naive implementation for Lemma 3 runs in $O(m|\mathbf{V}|^4 2^{|\mathbf{V}|})$, where $m{=}|\mathbb{P}|$ and $\ell{=}|\mathbb{F}|$. Further, if the problem instance is compatible with g-identifiability, where no partial-observability is involved, then the algorithm runs in a polynomial time in $|\mathbf{V}|$ since every identified MVEF will be a single-piece chunk and an exact cover algorithm will spot uncovered elements first.

5. Discussions: Completeness, Complexity, Conditional Effects, and Transportability

We conjecture that Lemma 3 is also a necessary condition and, hence, our algorithm is sound and complete. We can show, under the completeness conjecture, that the decision version of our problem is NP-complete. Further, we discuss two possible extensions to this work.

On Completeness Our intuition for its completeness lies in the fact that each gc-factor in $\mathbb{F}$ provides non-decomposable information about the model. Formally speaking, a probability $P_{\mathbf{w}}(\mathbf{z}) \in \mathbb{F}$ is not identifiable from the combination of smaller pieces, i.e., $\mathbb{P} = \{P_{\mathbf{W}'}(\mathbf{Z}') \mid \mathbf{Z}' \subseteq \mathbf{Z}, \mathbf{W}' \subseteq \mathbf{W}\} \setminus \{P_{\mathbf{W}}(\mathbf{Z})\}$. Following Lemma 2, for each

Algorithm 1 GID-PO

Input: $\mathcal{G}$ a causal graph, $\mathbf{x}$ and $\mathbf{y}$ value assignments for a query, $\mathbb{P}$ a collection of available distributions
Prepare $\mathbb{J}$ an empty collection.
for $\langle P_{\mathbf{Z}'}(\mathbf{V}'), \langle \mathbf{X}_i, \mathbf{Y}_i \rangle \rangle \in \mathbb{P} \times \mathbb{F}$ **do**
  continue if an MVEF that embeds $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$ is already found by the same data $P_{\mathbf{Z}'}(\mathbf{V}')$.
  $\langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle, \mathbf{W}^\dagger \leftarrow \text{MVEF}(\langle \mathbf{X}_i, \mathbf{Y}_i \rangle, P_{\mathbf{Z}'}(\mathbf{V}'))$.
  continue if the same MVEF $\langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle$ is already recorded in $\mathbb{J}$ from some data.
  if $\langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle$ exists **and** identifiable with $P_{\mathbf{Z}'}(\mathbf{V}')$ **then**
    $\mathbf{i} \leftarrow$ the indices of factors in $\mathbb{F}$ embedded in the MVEF.
    Add $\langle \langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle, \mathbf{W}^\dagger, \mathbf{i}, \langle \mathbf{Z}', \mathbf{V}' \rangle \rangle$ to $\mathbb{J}$.
  end if
end for
Let $\mathbb{I}$ be the collection of indices in $\mathbb{J}$.
if $\mathbb{I}' \leftarrow$exact cover with $\{1, \ldots, |\mathbb{F}|\}$ and $\mathbb{I}$ **then**
  Let $\mathbb{J}' \subseteq \mathbb{J}$ be the subcollection matching the solution $\mathbb{I}'$.
  Let $\mathbf{W}'$ be the union of all MVEF-admissible sets in $\mathbb{J}'$.
  return $\sum_{\mathbf{y}^+ \setminus (\mathbf{Y} \cup \mathbf{W}')} \prod_{\mathbb{J}'} \text{ID}(P_{\mathbf{x}^\dagger \setminus \mathbf{Z}'}(\mathbf{y}^\dagger), (\mathcal{G} \setminus \mathbf{Z}')\langle \mathbf{V}' \cap \mathbf{V}^+ \rangle, P_{\mathbf{z}'})$.
end if
return NULL.

gc-factor one can construct two models $\mathcal{M}^1$ and $\mathcal{M}^2$ agreeing on every distribution in $\mathbb{P}$ but differing on $P_{\mathbf{W}}(\mathbf{Z})$. Choose a bidirected spanning tree T on $\mathbf{Z}$ and assign each $W \in \mathbf{W}$ to exactly one of its children $c(W) \in \mathbf{Z}$. Let the UCs on the edges of T and the variables in $\mathbf{W}$ be independent fair coins, and give each $Z \in \mathbf{Z}$ an independent noise U_Z with $P(U_Z = 1) = \epsilon$, $0 < \epsilon < 0.5$. In $\mathcal{M}^1$, each Z is the exclusive-or of the UCs of its incident tree edges, the treatments assigned to it, and U_Z, ignoring its other parents; $\mathcal{M}^2$ flips the output of one $Z' \in \mathbf{Z}$; every other endogenous variable has the same nondegenerate mechanism in both models. Writing $r = |\mathbf{Z}|$, $P^{1,2}_{\mathbf{w}}(\mathbf{z}) = 2^{-r}\left[1 \pm (-1)^{\bigoplus \mathbf{z} \oplus \bigoplus \mathbf{w}}(1-2\epsilon)^r\right]$, so both models are strictly positive as Lemma 2 requires, every proper marginal of $\mathbf{Z}$ is uniform, and leaving any W unintervened averages the parity term away; hence all distributions in $\mathbb{P}$ agree while $P_{\mathbf{W}}(\mathbf{Z})$ differs. Further, estimates for proper embedding factors and other gc-factors will not pinpoint a gc-factor of interest. For example, does knowing a chunk $\{f, g\}$ and piece $\{g\}$ allow us to infer what $\{f\}$ is? Consider an equation $h = \int_x f(x) \cdot g(x) \mathrm{d}x$. Generally, knowing h and g does not allow us to infer what f is. Having a number of such equations may help characterizing f but it won't identify f unless the domain of X is small and finite. For such reasons, we conjecture that $\mathbb{F}$ provides a fundamental, necessary, and sufficient building blocks for constructing $P_{\mathbf{x}}(\mathbf{y})$ given marginal interventional distributions. A rigorous proof for the completeness is still under investigation.

On NP-Completeness Assume that we have proved the completeness of the algorithm, which reflects Prop. 2. Then, we can show that the decision problem of the puzzle is indeed NP-complete. We present a polynomial reduction of an exact cover problem to a GID-PO problem instance.

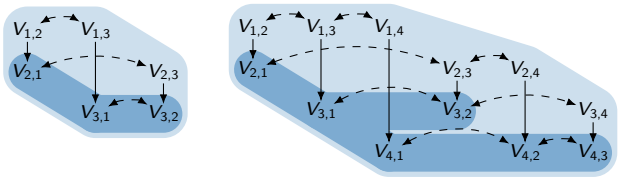
Consider an arbitrary exact cover problem with a universe $[k] = \{1, \ldots, k\}$ and a collection of its subsets.[6] We construct a causal graph $\mathcal{G}$ with $\mathbf{V} = \{V_{i,j}\}_{1 \le i \ne j \le k}$ and its direct edges are $V_{i,j} \to V_{j,i}$ for $i < j$. We add bidirected edges to connect $\mathbf{V}_{i,:} = \{V_{i,j}\}_{j \ne i}$ side by side. Examples of such causal graph with $k = 3$ and $k = 4$ are shown in Figs. 10a and 10b. We then fix a query as $P(\mathbf{V}')$ where $\mathbf{V}' = \{V_{j,i}\}_{j > i}$ are the sink nodes in $\mathcal{G}$. Each gc-factor $\langle pa(\mathbf{V}_{i,:}), \mathbf{V}_{i,:} \rangle \in \mathbb{F}$ represents element i in the universe $[k]$, e.g., $P(V_{1,2}, V_{1,3})$ for 1 and $P(V_{3,1}, V_{3,2} | do(V_{1,3}, V_{2,3}))$ for 3. Given a subset of integers $\mathbf{w} \subseteq [k]$, there exists a corresponding distribution $P_{\mathbf{X}^\dagger}(\mathbf{Y}^\dagger)$ in $\mathbb{P}$ such that $\langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle \in \mathbb{F}^\dagger$ is an MVEF of any of gc-factor corresponding to $w \in \mathbf{w}$ where $\mathbf{V}^\dagger = \mathbf{V} \setminus \{V_{i,j}\}_{\{i<j \,|\, i,j \in \mathbf{w}\}}$. For instance, a subset of integers for an exact cover problem in Fig. 10a will be matched to the following distributions:

$$
\begin{aligned}
\{1\} &= P(V_{1,2}, V_{1,3}) \\
\{2\} &= P(V_{2,1}, V_{2,3} \mid do(V_{1,2})) \\
\{3\} &= P(V_{3,1}, V_{3,2} \mid do(V_{1,3}, V_{2,3})) \\
\{1,2\} &= P(V_{1,3}, V_{2,1}, V_{2,3}) \\
\{1,3\} &= P(V_{1,2}, V_{3,1}, V_{3,2} \mid do(V_{2,3})) \\
\{2,3\} &= P(V_{2,1}, V_{3,1}, V_{3,2} \mid do(V_{1,2}, V_{1,3})) \\
\{1,2,3\} &= P(V_{2,1}, V_{3,1}, V_{3,2})
\end{aligned}
$$

Hence, the construction can be done in a polynomial time of $k|\mathbb{I}|$, the size of the problem. Further, Alg. 1 yields an identification formula if and only if the reduced problem has an exact cover solution since each set as an available distribution can only identify itself but not others. Therefore, under the completeness of Lemma 3, we can show that the decision problem of general identifiability under partial-observability is NP-hard. It is moreover in NP — hence NP-complete — because a polynomial-size certificate, namely a level $\mathbf{V}'$ with $\mathbf{V}^* \subseteq \mathbf{V}' \subseteq \mathbf{V}^+$ together with, for each gc-factor in $\mathbb{F}'$, a distribution in $\mathbb{P}$ identifying it, can be verified in polynomial time by Cor. 1.

Generalization to Transportability and Conditional Distributions One of the natural extensions of this work is in the context of *transportability* (Pearl & Bareinboim, 2011; Bareinboim & Pearl, 2014; Lee et al., 2020). Transportability is concerned with the fusion of datasets collected from heterogeneous domains, where the mechanisms for some of the variables differ from a domain in which a causal effect is sought. Unless the differences between a source and a target (pieces' shapes are incompatible) are on the $\mathbf{Y}$-side of a factor or on the variables projected out between its $\mathbf{X}$-

[6] An empty set can be simply ignored because it does not contribute to answering the exact cover problem.



(a) (b)

Figure 10. Causal diagrams for exact cover problems with universe (a) $\{1, 2, 3\}$ and (b) $\{1, 2, 3, 4\}$.

and $\mathbf{Y}$-sides — formally, unless $An(\mathbf{Y}^\dagger)_{\mathcal{G}_{\underline{\mathbf{X}^\dagger}}}$ intersects the discrepancies — a similar procedure can be applied.

Another generalization is identifying a *conditional* interventional distribution, e.g., $P_{\mathbf{x}}(\mathbf{y}|\mathbf{w})$ is delegated to identifying $P_{\mathbf{x},\mathbf{w}\setminus\mathbf{w}'}(\mathbf{y} \mid \mathbf{w}')$ for a suitable $\mathbf{w}' \subseteq \mathbf{w}$ (Tian, 2004; Lee et al., 2020). This change only requires an additional pre- and post-process of the result from GID-PO or its extension to transportability. We provide the generalization of our problem taking account both directions in Sec. F.

6. Conclusions

We introduced the general identifiability problem when the available distributions are only partially observable, which is named GID-PO. We investigated how a causal query can be factorized under different levels of projections, and then introduced new constructs called *embedding factors* and *co-identification*. These constructs make explicit the connection of the factors required to identify the targeted query and the available observed distributions, which allows a systematic view of the problem of identifiability under different granularities. We introduced a new graphical structure called *minimum viable embedding factor* (MVEF) and studied its properties, including its uniqueness, disjointedness, and compositionality. Putting these results together, we developed a new algorithm (GID-PO) that efficiently and systematically examines the identifiability of embedding factors and combines the identified MVEFs to compose the expression for a given query. Since each of the factors cannot be identified from smaller marginal interventional distributions, we conjecture that the procedure is also necessary. Assuming its completeness, we showed that the decision version of this new identifiability problem is NP-complete; yet, it does run in a polynomial time in the number of observed variables for the problems under full-observability (Lee et al., 2019). Noting that, in practice, available datasets are measured inconsistently with respect to the variables they cover – they usually have different columns – we hope the results in this paper can help data scientists tackle more challenging identification instances and determine causal effects in more intricate and realistic scenarios.

Acknowledgements

This research is supported in parts by grants from NSF (IIS-1704352 and IIS-1750807 (CAREER)).

A Supplementary Material for "Causal Effect Identifiability under Partial-Observability"

A. Latent Projection

We first provide a few remarks on latent projection.

Remark 1. $\mathcal{G}\langle\text{-}W\rangle$ subsumes $\mathcal{G}[\mathbf{V} \setminus \{W\}]$ and includes the following edges:

1. $\{Q \to C \mid Q \to W \to C \in \mathcal{G}\}$,
2. $\{S \leftrightarrow C \mid S \leftrightarrow W \to C \in \mathcal{G}, S \neq C\}$, and
3. $\{C \leftrightarrow C' \mid C \leftarrow W \to C' \in \mathcal{G}, C \neq C'\}$.

Remark 2. For any $Q, R \in \mathbf{V} \setminus \mathbf{W}$, if Q is an ancestor of R in $\mathcal{G}$, then Q is an ancestor of R in $\mathcal{G}\langle\text{-}\mathbf{W}\rangle$.

Remark 3. Let $\mathbf{W}$ be a set of variables in $\mathcal{G}$ such that $ch(W)_{\mathcal{G}} \neq \emptyset$ for every $W \in \mathbf{W}$. For any $Q, R \in \mathbf{V} \setminus \mathbf{W}$, if Q and R are connected via bidirected edges in $\mathcal{G}$, then so are they in $\mathcal{G}\langle\text{-}\mathbf{W}\rangle$.

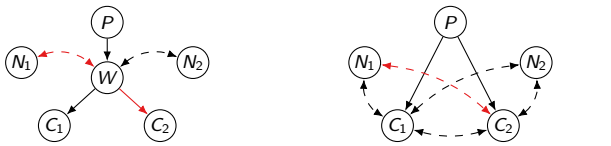
Fig. 11 illustrates the above remarks. For example, $P \to W \to C_1$ yields $P \to C_1$, $N_1 \leftrightarrow W \to C_2$ leaves $N_1 \leftrightarrow C_2$, and $C_1 \leftarrow W \to C_2$ leads to $C_1 \leftrightarrow C_2$ (Remark 1). C_1 and C_2 inherit ancestors of W by adopting its parent P as their parents (Remark 2). N_1 and N_2 are still connected via bidirected edges after projecting out W, which is not a sink (Remark 3).

B. Identifiability with Single Partially-Observed Observational Distribution

In this section, we study a simplified identifiability problem where the only available distribution is observational, which does not necessarily include all the modeled variables $\mathbf{V}$ of a given causal graph $\mathcal{G}$, hence, partially-observed. We first formally define the problem as follows:

Definition 9 (Causal Effect Identifiability under a Partially-observed Observational Distribution)**.** Given a causal graph $\mathcal{G}$, let $\mathbf{V}'$ be a subset of its vertices $\mathbf{V}$, and $\mathbf{X}$ and $\mathbf{Y}$ be disjoint subsets of $\mathbf{V}$. The causal effect $P_{\mathbf{x}}(\mathbf{y})$ is said to be identifiable from $P(\mathbf{V}')$ in $\mathcal{G}$ if $P_{\mathbf{x}}(\mathbf{y})$ is (uniquely) computable from $P(\mathbf{V}')$ in any causal model which induces $\mathcal{G}$ and $P(\mathbf{V}) > 0$.

Trivially, one can translate this setting into a classic identifiability problem where $\mathbf{U} \cup (\mathbf{V} \setminus \mathbf{V}')$ are unobserved and $\mathbf{V}'$ are observed variables. That is, a sound result can be obtained over the projection of $\mathcal{G}$ onto $\mathbf{V}'$ as follows.



(a) (b)

Figure 11. (a) A part of causal graph representing W and its neighbors and edges with one end on W. (b) Projecting out W creates new relationships among neighboring variables. Edges are colored to help understanding how projection works.

Proposition 6. *Let $\mathbf{V}' \subseteq \mathbf{V}$ satisfy $\mathbf{V}^* \subseteq \mathbf{V}'$. $P_{\mathbf{x}}(\mathbf{y})$ is identifiable from $P(\mathbf{V}')$ in $\mathcal{G}$ if $P_{\mathbf{x}^*}(\mathbf{y}^*)$ is identifiable from $P(\mathbf{V}')$ in $\mathcal{G}\langle\mathbf{V}'\rangle$.*

Proof. It follows from the equality $P_{\mathbf{x}^*}(\mathbf{y}^*) = P_{\mathbf{x}}(\mathbf{y})$ and the fact that $P_{\mathbf{x}^*}(\mathbf{y}^*)$ is identified with $P(\mathbf{V}')$. □

Still, it is not immediate to show its completeness. Causal knowledge over unseen variables might constrain the range of a causal effect so that it is identifiable. Non-identifiability can be characterized as follows.

Lemma 6. *$P_{\mathbf{x}}(\mathbf{y})$ is not identifiable from $P(\mathbf{V}')$ given $\mathcal{G}$ if there exist two models $\mathcal{M}^1$ and $\mathcal{M}^2$ compatible with $\mathcal{G}$ yielding $P^1(\mathbf{V}), P^2(\mathbf{V}) > 0$ and $P^1(\mathbf{V}') = P^2(\mathbf{V}')$ and $P^1_{\mathbf{x}}(\mathbf{y}) \neq P^2_{\mathbf{x}}(\mathbf{y})$.*

Proof. The existence of two models excludes the existence of a function mapping from $P(\mathbf{V}')$ to $P_{\mathbf{x}}(\mathbf{y})$. □

We begin to describe conditions necessary to identify a causal effect under partial-observability.

Proposition 7 (Outcome Inclusion Criterion)**.** *If $\mathbf{Y}^* \not\subseteq \mathbf{V}'$, then $P_{\mathbf{x}}(\mathbf{y})$ is not identifiable from $P(\mathbf{V}')$ in $\mathcal{G}$.*

Proof. Pick $Y \in \mathbf{Y}^* = \mathbf{Y}$ that is not in $\mathbf{V}'$. Define two models $\mathcal{M}^1$ and $\mathcal{M}^2$ of binary variables, compatible with $\mathcal{G}$, which share the same joint distributions for the exogenous variables *other than* U_Y, and the same functions for the endogenous variables excluding Y, where the functions for the children of Y are set to ignore Y. For Y, simply let $f^1_Y = U_Y$ and $f^2_Y = U_Y$ where $P^1(U_Y) \neq P^2(U_Y)$. By construction, two models share $P(\mathbf{V}')$ while disagreeing on $P(y)$, that is $P^1(y) \neq P^2(y)$. Since $P_{\mathbf{x}}(y) = P(y)$ in both models, they disagree on $P_{\mathbf{x}}(y)$, thus, on $P_{\mathbf{x}}(\mathbf{y})$, as well. □

We similarly establish a proposition for the treatment variables $\mathbf{X}$, however, with only a subset of them.

Proposition 8 (Minimal Treatment Inclusion Criterion)**.** *If $\mathbf{X}^* \not\subseteq \mathbf{V}'$, then $P_{\mathbf{x}}(\mathbf{y})$ is not identifiable from $P(\mathbf{V}')$ in $\mathcal{G}$.*

Proof. Pick $X \in \mathbf{X}^* \setminus \mathbf{V}'$. There must be at least one $Y \in \mathbf{Y}$ such that $X \in An(Y)_{\mathcal{G}_{\overline{\mathbf{X}}}}$, which implies the existence of a directed path from X to Y. Let the sequence of variables in the path be $\mathbf{P} = (X, V_1, \ldots, V_m)$ where $V_m = Y$. Let

$$f^1_X = U_X \qquad f^2_X = 1 - U_X$$
$$f^1_{V_1} = X \oplus U_1 \qquad f^2_{V_1} = (1 - X) \oplus U_1$$

where $P(U_X = 1) = P(U_1 = 1) = \epsilon$. For $2 \leq i \leq m$, define $f_{V_i} = V_{i-1} \oplus U_i$ where $P(U_i = 1) = \epsilon$.

Every other variable — in particular every off-path child of X that lies in $\mathbf{V}'$ — is given the same function in both models, chosen so as to ignore X (as in the proof of Prop. 7); otherwise it would inherit the two different marginals of X and break the required agreement. This guarantees that $P^1(\mathbf{P} \setminus \{X\}) = P^2(\mathbf{P} \setminus \{X\})$ and hence $P^1(\mathbf{V} \setminus \{X\}) = P^2(\mathbf{V} \setminus \{X\})$, so that $P^1(\mathbf{V}') = P^2(\mathbf{V}')$. Given $\epsilon \approx 0$, $P^1_{X=0}(Y = 1) \approx 0$ while $P^2_{X=0}(Y = 1) \approx 1$, which demonstrates the non-identifiability of $P_{\mathbf{x}}(\mathbf{y})$. □

Two necessary conditions imply that $\mathbf{V}'$ should include $\mathbf{V}^*$: $\mathbf{Y}$ and a subset of $\mathbf{X}$ that affect $\mathbf{Y}$ under the intervention on $\mathbf{X}$. Now we move to a sufficient condition.

We illustrate two examples demonstrating that the non-identifiability in the projection is equivalent to the non-identifiability under partial-observability. Consider a minimal, non-identifiable example in Fig. 12a.

Example 1. $P_x(y)$ is not identifiable from $P(X, Y)$ given a graph in Fig. 12b. Define two models with binary variables:

$$f_X = U,$$
$$f_Z = X \oplus U_Z,$$
$$f^1_Y = Z \oplus U \oplus U_\delta, \qquad f^2_Y = U_Y$$

where $P(U = 1) = 0.5$, $P(U_Z = 1) = \epsilon$, $P(U_\delta = 1) = \delta$ with $0 < \epsilon, \delta < 0.5$, and $P(U_Y = 1) = r := \epsilon + \delta - 2\epsilon\delta$. The private noise U_δ is what keeps the full joint strictly positive, as Def. 9 requires: without it $Y = X \oplus Z$ holds deterministically and half of the cells of $P(X, Z, Y)$ vanish. The two models yield the same observational distribution $P(x, y) = 0.5(yr + (1 - y)(1 - r)) > 0$ but $P^1_x(y) = 0.5$ and $P^2_x(y = 1) = r$. At $\epsilon = \delta = 0.1$ these are 0.5 and 0.18.

Example 2. $P_x(y)$ is not identifiable from $P(X, Y)$ given a graph in Fig. 12c. Define two binary causal models:

$$f_Z = U_Z,$$
$$f_X = Z \oplus U_X,$$
$$f^1_Y = X \oplus Z \oplus U_\delta, \qquad f^2_Y = U_Y$$

where $P(U_Z = 1) = 0.5$, $P(U_X = 1) = \epsilon$, $P(U_\delta = 1) = \delta$ with $0 < \epsilon, \delta < 0.5$, and $P(U_Y = 1) = r := \epsilon + \delta - 2\epsilon\delta$. Both models produce $P(x, y) = 0.5(yr + (1-y)(1-r)) > 0$ but $P^1_x(y) = 0.5$ and $P^2_x(y = 1) = r$. At $\epsilon = \delta = 0.1$ these are 0.5 and 0.18.

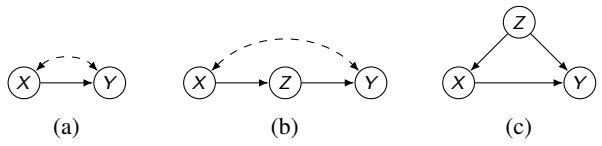


Figure 12. The first graph (a) known as a bow-arc graph is the result of projecting out Z from (b) and (c), respectively. Given $P(X, Y, Z)$, $P_x(y)$ is identifiable from (b) and (c).

A formal parametrization of two models is given for a hedge (Shpitser & Pearl, 2006a) and for mapping the root set of the hedge to a subset of $\mathbf{Y}$ (Lee et al., 2019). We construct two models on $\mathcal{G}$ that realize, under the projection onto $\mathbf{V}'$, a hedge $\langle \mathcal{F}, \mathcal{F}' \rangle$ for $P_{\mathbf{x}^*}(\mathbf{y})$ in $\mathcal{G}'$, such that both models are strictly positive on $\mathbf{V}$, agree on $P(\mathbf{V}')$, and differ on $P^1_{\mathbf{x}^*}(\mathbf{y}) \neq P^2_{\mathbf{x}^*}(\mathbf{y})$.

Lemma 7 (Hedge Exclusion)**.** *$P_{\mathbf{x}}(\mathbf{y})$ is not identifiable from $P(\mathbf{V}')$ in $\mathcal{G}$ if there exists a hedge for $P_{\mathbf{x}^*}(\mathbf{y})$ in $\mathcal{G}'$.*

Proof. Throughout, $\mathcal{G}' = \mathcal{G}\langle \mathbf{V}' \rangle$ and $\langle \mathcal{F}, \mathcal{F}' \rangle$ is an $\mathbf{R}$-rooted hedge for $P_{\mathbf{x}^*}(\mathbf{y})$ in $\mathcal{G}'$: $\mathcal{F}' \subseteq \mathcal{F}$ are $\mathbf{R}$-rooted c-forests, $\mathcal{F} \cap \mathbf{X}^* \neq \emptyset$, $\mathcal{F}' \cap \mathbf{X}^* = \emptyset$, and every $R \in \mathbf{R}$ has a directed path to $\mathbf{Y}$ in $\mathcal{G}'$ with no internal vertex in $\mathbf{X}^*$. We use the following structural facts. (C1) The bidirected edges of $\mathcal{F}$ connect $\mathbf{V}_{\mathcal{F}}$, and those of $\mathcal{F}'$ connect $\mathbf{V}_{\mathcal{F}'}$. (C2) Every vertex of $\mathcal{F}$ has at most one child in $\mathcal{F}$, and $\mathbf{R}$ is the set of vertices without a child; hence each $a \in \mathbf{V}_{\mathcal{F}}$ has a unique descending chain $a, \mathrm{ch}(a), \mathrm{ch}^2(a), \ldots$ ending at a root. (C3) No directed edge of $\mathcal{F}$ leaves $\mathcal{F}'$: if $f' \in \mathcal{F}'$ had its child outside $\mathcal{F}'$, then f' would have no child in $\mathcal{F}'$ and would belong to $\mathbf{R}$, but roots have no child in $\mathcal{F}$ either. Thus $\mathcal{F}'$ is closed under taking children, and every vertex of $\mathcal{F} \setminus \mathcal{F}'$ has all its ancestors in $\mathcal{F}$ inside $\mathcal{F} \setminus \mathcal{F}'$. (C4) $\mathcal{F} \setminus \mathcal{F}' \neq \emptyset$, every vertex of $\mathcal{F} \setminus \mathcal{F}'$ has a child, and $\mathbf{R} \cap \mathbf{X}^* = \emptyset$.

Since $\mathbf{X} \setminus \mathbf{X}^*$ are not ancestors of $\mathbf{Y}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$, rule 3 of do-calculus gives $P_{\mathbf{x}}(\mathbf{y}) = P_{\mathbf{x}^*}(\mathbf{y})$ in every model compatible with $\mathcal{G}$. By Lemma 6 it therefore suffices to construct two models $\mathcal{M}^1$ and $\mathcal{M}^2$ compatible with $\mathcal{G}$ such that $P^1(\mathbf{V}), P^2(\mathbf{V}) > 0$, $P^1(\mathbf{V}') = P^2(\mathbf{V}')$ and $P^1_{\mathbf{x}^*}(\mathbf{y}) \neq P^2_{\mathbf{x}^*}(\mathbf{y})$.

Linear models. Every variable is a tuple of bits (*coordinates*); a mechanism may read any coordinates of its parents and may ignore the others. Every mechanism below is the exclusive-or of some inputs, some fair coins U (one per bidirected edge used) and independent private noises $N \sim \mathrm{Bernoulli}(\epsilon_N)$ with $0 < \epsilon_N < \frac{1}{2}$. For a set S of coordinates let $\chi_S := \bigoplus_{v \in S} v$; it is an affine function of the exogenous bits, and a bit is said to *enter* χ_S if it appears an odd number of times in this expansion. Then

$$\mathbb{E}\big[(-1)^{\chi_S}\big] = \pm\, 1[\text{no fair coin enters } \chi_S] \times \prod_{N \text{ enters } \chi_S} (1 - 2\epsilon_N), \tag{5}$$

and the joint distribution of any set of coordinates is determined by these expectations over all its subsets S.

Realizing the edges of $\mathcal{F}$ in $\mathcal{G}$. For each directed edge $A \to B$ of $\mathcal{F}$ that is not an edge of $\mathcal{G}$, fix a directed path $A = w_0 \to w_1 \to \cdots \to w_k \to B$ in $\mathcal{G}$ with $w_1, \ldots, w_k \notin \mathbf{V}'$, give each w_i a coordinate $w_i^{(A\to B)} = w_{i-1}^{(A\to B)} \oplus N$ (where $w_0^{(A\to B)}$ is the coordinate A defined below), and let B read $w_k^{(A\to B)}$. Then B receives $A \oplus \eta_{A\to B}$ with $\eta_{A\to B}$ an accumulated noise independent of all other primitive exogenous bits ($\eta_{A\to B} = 0$ if $A \to B$ is an edge of $\mathcal{G}$). For each bidirected edge $A \leftrightarrow B$ of $\mathcal{F}$ fix a fresh fair coin U_{AB}. If $A \leftrightarrow B$ is an edge of $\mathcal{G}$, both endpoints read U_{AB}. Otherwise the definition of projection provides either a vertex $W \notin \mathbf{V}'$ with directed paths to A and to B meeting only at W, or a bidirected edge $W_1 \leftrightarrow W_2$ of $\mathcal{G}$ with (possibly trivial) directed paths $W_1 \to \cdots \to A$ and $W_2 \to \cdots \to B$ whose internal vertices are outside $\mathbf{V}'$; give W the coordinate $W^{(AB)} = U_{AB}$, or give W_1 and W_2 (when they differ from A and B) the coordinates $U_{AB} \oplus N$ and $U_{AB} \oplus N'$, and transmit along the paths by noisy copies as above. Either way A receives $U_{AB} \oplus \eta_{AB,A}$ and B receives $U_{AB} \oplus \eta_{AB,B}$ with independent accumulated noises. A vertex or a bidirected edge of $\mathcal{G}$ serving several edges of $\mathcal{F}$ carries one coordinate (one coin) per edge served. These coordinates and mechanisms are the same in $\mathcal{M}^1$ and $\mathcal{M}^2$.

The hedge coordinates. Each $v \in \mathbf{V}_{\mathcal{F}}$ has a binary coordinate, also denoted v, with a private noise N_v. Let $\mathcal{E}_{\mathcal{F}}$ be the set of bidirected edges of $\mathcal{F}$ and $pa(v)_{\mathcal{F}}$ the parents of v in $\mathcal{F}$. In $\mathcal{M}^1$,

$$v = \bigoplus_{p \in pa(v)_{\mathcal{F}}} \big(p \oplus \eta_{p\to v}\big) \;\oplus \bigoplus_{e \in \mathcal{E}_{\mathcal{F}}:\, v \in e} \big(U_e \oplus \eta_{e,v}\big) \;\oplus N_v,$$

where all other parents of v in $\mathcal{G}$ and all other confounders are ignored. In $\mathcal{M}^2$ the vertices of $\mathcal{F} \setminus \mathcal{F}'$ are as in $\mathcal{M}^1$, while a vertex $v \in \mathbf{V}_{\mathcal{F}'}$ reads only its parents in $\mathcal{F}'$ and the bidirected edges of $\mathcal{F}'$:

$$v = \bigoplus_{p \in pa(v)_{\mathcal{F}'}} \big(p \oplus \eta_{p\to v}\big) \;\oplus \bigoplus_{e \in \mathcal{E}_{\mathcal{F}'}:\, v \in e} \big(U_e \oplus \eta_{e,v}\big) \;\oplus N_v \;\oplus 1[v = r_0]\, N_\delta,$$

where $r_0 \in \mathbf{R}$ is a fixed root and $N_\delta \sim \mathrm{Bernoulli}(\delta)$ is a noise present only in $\mathcal{M}^2$, with δ fixed below. The two models share every other exogenous variable and every noise term; they differ only in which inputs the vertices of $\mathcal{F}'$ read, and in N_δ.

Transfer to $\mathbf{Y}$ *and the remaining variables.* For each $r \in \mathbf{R} \setminus \mathbf{Y}$ fix a directed path π_r in $\mathcal{G}'$ from r to some $y_r \in \mathbf{Y}$ whose internal vertices avoid $\mathbf{X}^*$; realize each of its edges in $\mathcal{G}$ as above and give every vertex on the realized path a coordinate that is a noisy copy of the previous one, starting from the hedge coordinate r. Then y_r carries a coordinate equal to $r \oplus \eta_r$ with η_r independent of all other primitive exogenous bits. Every coordinate not introduced so far, observed or not, is an independent fair coin. The coordinates of this paragraph have the same mechanisms in both models.

Positivity and compatibility. Every endogenous coordinate is the exclusive-or of earlier coordinates and a private exogenous bit (its own noise or its own coin) read by no other coordinate; proceeding in topological order, every joint value of all coordinates has positive probability, so $P^1(\mathbf{V}), P^2(\mathbf{V}) > 0$. Only edges of $\mathcal{G}$ and confounders of bidirected edges of $\mathcal{G}$ are used, so both models are compatible with $\mathcal{G}$.

Observational agreement. Write ξ_a for the exogenous input of the hedge coordinate a in $\mathcal{M}^1$ (its coins and all its noise terms). By (C2), in $\mathcal{M}^1$ every hedge coordinate satisfies $v = \bigoplus_{a \in An(v)_{\mathcal{F}} \cup \{v\}} \xi_a$, because the input of an ancestor reaches v along its unique chain exactly once. Hence, for $S \subseteq \mathbf{V}_{\mathcal{F}}$, $\chi_S = \bigoplus_a m_S(a)\, \xi_a$ with $m_S(a) := |S \cap \mathrm{chain}(a)| \bmod 2$, and the coin of an edge $\{a, b\}$ enters χ_S iff $m_S(a) \neq m_S(b)$. By (C1), $\mathbb{E}^1[(-1)^{\chi_S}] \neq 0$ requires m_S to be constant on $\mathbf{V}_{\mathcal{F}}$. Since $m_S(r) = 1[r \in S]$ for $r \in \mathbf{R}$ and $m_S(a) = 1[a \in S] \oplus m_S(\mathrm{ch}(a))$ otherwise, induction along the chains shows that $m_S \equiv 0$ iff $S = \emptyset$ and $m_S \equiv 1$ iff $S = \mathbf{R}$. Thus the only nonzero expectations are at $S = \emptyset$ and $S = \mathbf{R}$. In $\chi_{\mathbf{R}} = \bigoplus_{a \in \mathbf{V}_{\mathcal{F}}} \xi_a$ every coin enters twice and cancels, and every noise attached to a vertex or an edge of $\mathcal{F}$ enters exactly once; let J^1 be this set of noises and $\gamma := \prod_{N \in J^1}(1 - 2\epsilon_N) > 0$. In $\mathcal{M}^2$ write $S = S' \cup S''$ with $S' = S \cap \mathbf{V}_{\mathcal{F}'}$ and $S'' = S \setminus \mathbf{V}_{\mathcal{F}'}$. By (C3), the hedge coordinates of $\mathcal{F} \setminus \mathcal{F}'$ involve only inputs of $\mathcal{F} \setminus \mathcal{F}'$, and those of $\mathcal{F}'$ only inputs of $\mathcal{F}'$ along chains inside $\mathcal{F}'$. The coin of an edge $\{a, b\} \subseteq \mathbf{V}_{\mathcal{F}'}$ enters χ_S iff $m'_{S'}(a) \neq m'_{S'}(b)$, where m' is computed in $\mathcal{F}'$; the coin of an edge $\{a, b\} \subseteq \mathbf{V}_{\mathcal{F}} \setminus \mathbf{V}_{\mathcal{F}'}$ iff $m''_{S''}(a) \neq m''_{S''}(b)$; and the coin of an edge $\{a, b\}$ with $a \notin \mathbf{V}_{\mathcal{F}'}$ and $b \in \mathbf{V}_{\mathcal{F}'}$ iff $m''_{S''}(a) = 1$, since b does not read it. A nonzero expectation therefore requires $m'_{S'}$ constant on $\mathbf{V}_{\mathcal{F}'}$, i.e. $S' \in \{\emptyset, \mathbf{R}\}$ by the chain argument in $\mathcal{F}'$, and $m''_{S''} \equiv 0$ on $\mathbf{V}_{\mathcal{F}} \setminus \mathbf{V}_{\mathcal{F}'}$, because by (C1) every bidirected component of $\mathcal{F} \setminus \mathcal{F}'$ is joined to $\mathcal{F}'$ by an edge of $\mathcal{F}$. Since every vertex of $\mathcal{F} \setminus \mathcal{F}'$ has a child (C4), induction along the chains, starting from the vertices whose child lies in $\mathcal{F}'$, gives $S'' = \emptyset$. So in $\mathcal{M}^2$ too the only nonzero expectations are at $S = \emptyset$ and $S = \mathbf{R}$, and $\chi_{\mathbf{R}}$ is the exclusive-or of N_δ and of the noises attached to vertices and edges of $\mathcal{F}'$, a subset $J^2 \subseteq J^1$ because both models read the same noise terms wherever $\mathcal{M}^2$ reads an edge. Choosing

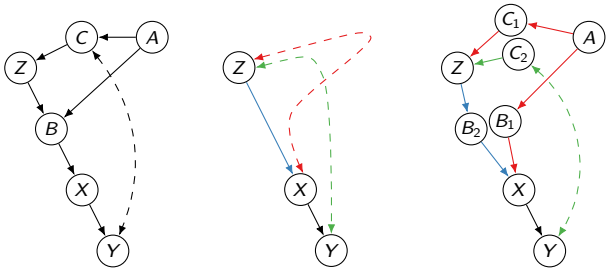
$$1 - 2\delta = \prod_{N \in J^1 \setminus J^2} (1 - 2\epsilon_N) \in (0, 1]$$

gives $\mathbb{E}^2[(-1)^{\chi_{\mathbf{R}}}] = \gamma = \mathbb{E}^1[(-1)^{\chi_{\mathbf{R}}}]$, hence $P^1(\mathbf{V}_{\mathcal{F}}) = P^2(\mathbf{V}_{\mathcal{F}})$ by (5). The remaining observed coordinates are transfer coordinates, which are functions of the hedge coordinates of $\mathbf{R}$ and of private noises through mechanisms shared by the two models, and independent fair coins; their conditional distribution given $\mathbf{V}_{\mathcal{F}}$ is the same in both models, so $P^1(\mathbf{V}') = P^2(\mathbf{V}')$.

Interventional disagreement. Under $do(\mathbf{x}^*)$ the hedge coordinates of $\mathcal{F}'$ in $\mathcal{M}^2$ depend, after substituting the lifted paths, only on inputs of $\mathcal{F}'$ and on lifting coordinates that avoid $\mathbf{X}^*$, and $\mathcal{F}' \cap \mathbf{X}^* = \emptyset$, so $P^2_{\mathbf{x}^*}(\mathbf{V}_{\mathcal{F}'}) = P^2(\mathbf{V}_{\mathcal{F}'})$ and $\mathbb{E}^2_{\mathbf{x}^*}[(-1)^{\chi_{\mathbf{R}}}] = \gamma$. In $\mathcal{M}^1$ let A_0 be the set of vertices $a \in \mathbf{V}_{\mathcal{F}} \setminus \mathbf{X}^*$ with no vertex of $\mathbf{X}^*$ strictly below a on its chain. After the intervention $\chi_{\mathbf{R}}$ equals $\bigoplus_{a \in A_0} \xi_a$ up to constants and noises, so the coin of an edge $\{a, b\}$ enters iff exactly one endpoint lies in A_0. As $\mathbf{R} \subseteq A_0$ and $\mathcal{F} \cap \mathbf{X}^* \subseteq \mathbf{V}_{\mathcal{F}} \setminus A_0 \neq \emptyset$, (C1) yields such an edge, whence $\mathbb{E}^1_{\mathbf{x}^*}[(-1)^{\chi_{\mathbf{R}}}] = 0$. Thus $P^1_{\mathbf{x}^*}(\chi_{\mathbf{R}} = 0) = \frac{1}{2}$ whereas $P^2_{\mathbf{x}^*}(\chi_{\mathbf{R}} = 0) = \frac{1}{2}(1 + \gamma) > \frac{1}{2}$. Finally, the paths π_r avoid $\mathbf{X}^*$, so under $do(\mathbf{x}^*)$ the parity of the coordinates carried by $\mathbf{Y}$ that originate in $\mathbf{R}$ (the transfer coordinates together with the hedge coordinates of $\mathbf{R} \cap \mathbf{Y}$) equals $\chi_{\mathbf{R}} \oplus \bigoplus_r \eta_r$ with the η_r mutually independent and independent of the hedge coordinates; its expectation is 0 in $\mathcal{M}^1$ and $\gamma \prod_r (1 - 2\epsilon_{\eta_r}) > 0$ in $\mathcal{M}^2$. Hence $P^1_{\mathbf{x}^*}(\mathbf{y}) \neq P^2_{\mathbf{x}^*}(\mathbf{y})$, which completes the proof. □

For an illustrative purpose, consider the causal graph in Fig. 13a, where $P_x(y)$ is not identifiable from $P(Z, X, Y)$. The projection of $\mathcal{G}$ onto $\mathbf{V}' = \{Z, X, Y\}$ includes one new directed edge $Z \to X$, which corresponds to $Z \to B \to X$, and two new bidirected edges $Z \leftrightarrow X$ and $Z \leftrightarrow Y$, which correspond to $Z \leftarrow C \leftarrow A \to B \to X$ and $Z \leftarrow C \leftrightarrow Y$, respectively, in $\mathcal{G}$; the hedge is $\mathcal{F} = \{Z, X, Y\}$ with $\mathbf{R} = \mathcal{F}' = \{Y\}$. In the construction of Lemma 7, A carries the fair coin U_{ZX}, U_{ZY} is the confounder of $C \leftrightarrow Y$, B carries one coordinate for $Z \to B \to X$ and one for the branch $A \to B \to X$, and C carries one coordinate for the branch $A \to C \to Z$ and one for $C \leftrightarrow Y$; the four coordinates of B and C and the hedge coordinates Z, X, Y each have an independent private noise, while the coordinate at A is the fair coin U_{ZX}. Then $f^1_Y = X \oplus U_{ZY} \oplus N_Y$ while $f^2_Y = N_Y \oplus N_\delta$. With every noise level equal to ϵ, the noises that $\mathcal{M}^2$ drops are N_Z, N_X and the four path noises, so $1 - 2\delta = (1 - 2\epsilon)^6$: at $\epsilon = 0.1$, $\delta = 0.368928$, and $P^1_x(Y{=}0) = 0.5$ whereas $P^2_x(Y{=}0) = \frac{1}{2}\big(1 + (1 - 2\epsilon)^7\big) = 0.6048576$.

Theorem 2. *$P_{\mathbf{x}}(\mathbf{y})$ is identifiable from $P(\mathbf{V}')$ in $\mathcal{G}$ if and only if (i) $\mathbf{X}^* \subseteq \mathbf{V}'$, (ii) $\mathbf{Y} \subseteq \mathbf{V}'$, and (iii) there exists no*



(a) $\mathcal{G}$ (b) $\mathcal{G}\langle\mathbf{V}'\rangle$ (c) modeling of $\mathcal{G}$

Figure 13. Construction of two models demonstrating the non-identifiability of a query makes use of separate bits to represent paths responsible for the newly created edges as the result of projection. (a) a causal graph, (b) the projection of the graph with newly added edges color-coded, and (c) a schematic graph for parametrization of two models.

hedge for $P_{\mathbf{x}^*}(\mathbf{y})$ *in* $\mathcal{G}'$.

Proof. (if) There exists a formula for $P_{\mathbf{x}^*}(\mathbf{y}) = P_{\mathbf{x}}(\mathbf{y})$ using only terms derivable from $P(\mathbf{V}')$. (only if) This follows from outcome inclusion (Prop. 7), minimal treatment inclusion (Prop. 8), and hedge exclusion (Lemma 7). □

C. Proofs for Sec. 3 (Identifiability with a Single Partially-Observed Distribution)

We provided omitted proofs.

Proposition 1. *A causal effect* $P_{\mathbf{x}}(\mathbf{y})$ *is identifiable from* $P_{\mathbf{Z}'}(\mathbf{V}')$ *in* $\mathcal{G}$ *only if*

- $\mathbf{Y} \subseteq \mathbf{V}'$ *(inclusion of outcomes),*
- $\mathbf{X}^* \subseteq \mathbf{V}' \cup \mathbf{Z}'$ *(inclusion of minimal treatments), and*
- $\mathbf{Z}' \cap \mathbf{Y}^+ = \emptyset$ *(undisturbed mechanisms).*

Proof. If the third criterion does not meet and, hence, the mechanisms are disturbed, let there exist a directed path in $\mathcal{G}$ from some $Z \in \mathbf{Z}'$ to $Y \in \mathbf{Y}$ and no other $\mathbf{Z}' \setminus \{Z\}$, $\mathbf{X}$, and $\mathbf{Y} \setminus \{Y\}$ appears in the path. Let $0 < P(U_Z = 1) < 0.5$ and $f_Z^1 = U_Z$ and $f_Z^2 = 1 - U_Z$. Let the variables in the path carry the value of their parent in the path with a small noise. All other variables are fair coins. Then, the two models agree on $P_{\mathbf{z}'}(\mathbf{v}')$ but $P^1_{\mathbf{x}}(Y = 1) < P^2_{\mathbf{x}}(Y = 1)$.

Otherwise if mechanisms are *undisturbed*, the proofs for the failure to satisfy the first two criteria follow from Props. 7 and 8 since no $\mathbf{Z}'$ involves in the construction of the counterexamples. □

Lemma 1. *Given* $P_{\mathbf{x}}(\mathbf{y})$ *and* $P_{\mathbf{Z}'}(\mathbf{V}')$ *in* $\mathcal{G}$ *satisfying the three criteria in Prop. 1,* $P_{\mathbf{x}}(\mathbf{y})$ *is identifiable from* $P_{\mathbf{Z}'}(\mathbf{V}')$ *in* $\mathcal{G}$ *if and only if* $Q_{\mathbf{x}^* \setminus \mathbf{z}'}(\mathbf{y})$ *is identifiable in* $(\mathcal{G} \setminus \mathbf{Z}')\langle \mathbf{V}' \cap \mathbf{V}^+ \rangle$ *where* $Q = P_{\mathbf{z}'}$ *for any* $\mathbf{z}'$ *that agrees with* $\mathbf{x}$ *on* $\mathbf{X}^* \cap \mathbf{Z}'$.

Proof. (Soundness) By the condition, there exists a formula g such that $Q_{\mathbf{x}^* \setminus \mathbf{z}'}(\mathbf{y}) = g(Q(\mathbf{V}' \cap \mathbf{V}^+))$. Since $Q_{\mathbf{x}^* \setminus \mathbf{z}'}(\mathbf{y}) = P_{\mathbf{x}}(\mathbf{y})$ by

$$\begin{aligned} Q_{\mathbf{x}^* \setminus \mathbf{z}'}(\mathbf{y}) &= P_{\mathbf{z}', \mathbf{x}^* \setminus \mathbf{z}'}(\mathbf{y}) \\ &= P_{\mathbf{x}^+ \cap \mathbf{z}', \mathbf{x}^* \setminus \mathbf{z}'}(\mathbf{y}) \\ &= P_{\mathbf{x}^+ \cap \mathbf{z}', \mathbf{x}^+ \setminus \mathbf{z}'}(\mathbf{y}) \\ &= P_{\mathbf{x}^+}(\mathbf{y}) \\ &= P_{\mathbf{x}}(\mathbf{y}) \end{aligned}$$

and

$$Q(\mathbf{v}' \cap \mathbf{v}^+) = P_{\mathbf{z}'}(\mathbf{v}' \cap \mathbf{v}^+) = \sum_{\mathbf{v}' \setminus \mathbf{v}^+} P_{\mathbf{z}'}(\mathbf{v}'),$$

$P_{\mathbf{x}}(\mathbf{y})$ is identifiable as $P_{\mathbf{x}}(\mathbf{y}) = g(P_{\mathbf{z}'}(\mathbf{V}' \cap \mathbf{V}^+))$.

(Completeness) By Lemma 7, we can construct a counterexample which do not involve any $\mathbf{Z}'$, then incorporate $\mathbf{Z}'$ as independent random variables which will be ignored by their children. □

D. Proofs for Sec. 4 (Identifiability with Multiple Partially-Observed Distributions)

We first show the derivation for $P_x(y)$ in Fig. 7a.

$$\begin{aligned} P_x(y) &= \sum_{a,b,c} P_x(a,b,c,y) \\ &= \sum_{a,b,c} P(a) P_{a,x}(b) P_b(c) P_c(y) \\ &= \sum_{a,c} P(a) \left(\sum_b P_{a,x}(b) P_b(c) \right) P_c(y) \\ &= \sum_{a,c} P(a) P_{a,x}(c) P_c(y) \\ &= \sum_{a,c} P_{x'}(a) P_x(c|a) \left(\sum_b P(y|b,c) P(b) \right) \end{aligned}$$

where x' is any X value. The first equality is based on the definition of marginal probability. The second equality comes from decomposition. The third one is due to algebra and the fourth one is based on Prop. 3. Finally, rule 3 of do-calculus and back-door criterion is used to yield the final expression where one may contract $P_{x'}(a)P_x(c|a) = P_x(a,c)$ assuming $x' = x$. This derivation is helpful to highlight the role of the decomposition based on the causal graph. Alternatively, one could perform the following decomposition

(without considering B at the beginning):

$$\begin{aligned} P_x(y) &= \sum_{a,c} P_x(a,c,y) \\ &= \sum_{a,c} P(a) P_{a,x}(c) P_c(y) \\ &= \sum_{a,c} P_{x'}(a) P_x(c|a) \left(\sum_b P(y|b,c) P(b) \right) \end{aligned}$$

Lemma 2. *Let $\mathcal{G}$ be a causal graph and $\mathbf{X}, \mathbf{Y} \subseteq \mathbf{V}$. A query $P_{\mathbf{x}}(\mathbf{y})$ is not identifiable if there exist two causal models $\mathcal{M}^1$ and $\mathcal{M}^2$ compatible with $\mathcal{G}$ such that $P^1_{\mathbf{Z}_i}(\mathbf{V}_i) = P^2_{\mathbf{Z}_i}(\mathbf{V}_i)$, for every $P_i \in \mathbb{P}$, but $P^1_{\mathbf{x}}(\mathbf{y}) \neq P^2_{\mathbf{x}}(\mathbf{y})$, $P^1_{\mathbf{Z}_i}(\mathbf{V} \setminus \mathbf{Z}_i) > 0$, and $P^2_{\mathbf{Z}_i}(\mathbf{V} \setminus \mathbf{Z}_i) > 0$.*

Proof. The existence of two models excludes the existence of a function mapping from the given distributions to $P_{\mathbf{x}}(\mathbf{y})$. □

Proposition 2 (gc-decomposition)**.** *For $\mathbf{V}^* \subseteq \mathbf{V}' \subseteq \mathbf{V}^+$,*

$$P_{\mathbf{x}}(\mathbf{y}) = \sum_{\mathbf{y}^{+\prime} \setminus \mathbf{Y}} \prod_{\{i | \langle \mathbf{X}_i, \mathbf{Y}_i \rangle \in \mathbb{F}'\}} P_{\mathbf{x}_i}(\mathbf{y}_i) \qquad (2)$$

Proof. The decomposition follows from (Shpitser & Pearl, 2006a) under the projection $\mathcal{G}'$ with an additional refinement on the intervention to keep those only relevant to $\mathbf{Y}_i$. □

Lemma 3 (Soundness)**.** *Let $\mathcal{G}$, $P_{\mathbf{x}}(\mathbf{y})$, $\mathbb{P}$ be the causal graph, the query, and the distributions forming a GID-PO instance. If there exists a subset of variables $\mathbf{V}^* \subseteq \mathbf{V}' \subseteq \mathbf{V}^+$ such that every term $P_{\mathbf{x}_i}(\mathbf{y}_i)$ in each gc-factor $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$ in $\mathbb{F}'$ is identifiable from $P \in \mathbb{P}$, then $P_{\mathbf{x}}(\mathbf{y})$ is identifiable from $\mathbb{P}$ in $\mathcal{G}$.*

Proof. This follows from Prop. 2. □

Proposition 9 (Factors under Singleton Marginalization)**.** *Let $W \in \mathbf{V}'$ and $\mathbf{V}'' = \mathbf{V}' \setminus \{W\}$ where $W \notin \mathbf{V}^*$. Then, gc-factors in $\mathbb{F}'$ that contains W are merged to yield $\langle \mathbf{X}''_k, \mathbf{Y}''_k \rangle \in \mathbb{F}''$ such that*

$$\mathbf{Y}''_k = \left(\cup_{j \in \mathbf{j}} \mathbf{Y}'_j\right) \setminus \{W\}, \quad \mathbf{X}''_k = (\cup_{j \in \mathbf{j}} \mathbf{X}'_j) \setminus \mathbf{Y}''_k \setminus \{W\},$$

where $\mathbf{j}$ is the set of indices such that $\langle \mathbf{X}'_j, \mathbf{Y}'_j \rangle \in \mathbb{F}'$ and $Ch(W)_{\mathcal{G}'} \cap \mathbf{Y}'_j \neq \emptyset$. Other gc-factors without W are remained intact appearing in both $\mathbb{F}'$ and $\mathbb{F}''$.

Proof. The factor with W as in its $\mathbf{Y}$-side is merged with the factors having W on their $\mathbf{X}$ sides (due to the 2nd and 3rd items in Remark 1). The $\mathbf{Y}$ side of the resulting factor will simply be the union of $\mathbf{Y}$ sides of the factors without W. The parents of those connected via bidirected edges will be the same except the fact that the variables having W as a parent will now inherit the parents of W, which is included in the $\mathbf{X}$ side of the factor having W in its $\mathbf{Y}$ side. Hence, the result follows. □

Proposition 3 (Factors under Marginalization)**.** *Let $\mathbf{W} \subset \mathbf{V}'$ and $\mathbf{V}'' = \mathbf{V}' \setminus \mathbf{W}$ where $\mathbf{W} \cap \mathbf{V}^* = \emptyset$. Let $\mathcal{H}$ be an undirected graph where vertices are $\langle \mathbf{X}'_j, \mathbf{Y}'_j \rangle \in \mathbb{F}'$ and an edge exists if two factors satisfy their $\mathbf{Y}$-sides intersecting with $Ch(W)_{\mathcal{G}'}$ for some $W \in \mathbf{W}$. Then, vertices (i.e., gc-factors) in each connected component of $\mathcal{H}$ are merged to form $\langle \mathbf{X}''_k, \mathbf{Y}''_k \rangle \in \mathbb{F}''$ such that*

$$\mathbf{Y}''_k = \left(\cup_{j \in \mathbf{j}} \mathbf{Y}'_j\right) \setminus \mathbf{W}, \quad \mathbf{X}''_k = (\cup_{j \in \mathbf{j}} \mathbf{X}'_j) \setminus \mathbf{Y}''_k \setminus \mathbf{W},$$

where $\mathbf{j}$ is the set of indices of gc-factors in $\mathbb{F}'$ forming the connected component. Other gc-factors without $\mathbf{W}$ are remained intact, shared by both $\mathbb{F}'$ and $\mathbb{F}''$.

Proof. Consider an arbitrarily sequence of $\mathbf{W}$ where $W \in \mathbf{W}$ is sequentially marginalized out. By the condition, factors will be combined only if they cover W. Hence, there will be a partition of gc-factors to be combined. How they are combined follows from Prop. 9. □

Algorithm for obtaining MVEF and MVEF-admissible set is depicted in Alg. 2. This checks the three criteria in an iterative manner. When a current factor does not satisfy inclusion criteria, whether it be minimal treatment or outcome, and it is due to the lack of non-essential variables (i.e., excluding $\mathbf{V}^* = \mathbf{X}^* \cup \mathbf{Y}^*$), then one must address the issue by projecting out those variables and obtain a higher-level of embedding factor. However, if it is due to the absence of essential variables $\mathbf{V}^*$ or disturbed mechanisms, then no further projection is investigated since none of its embedding factors will satisfy those criteria. The soundness of the procedure follows from Prop. 3 that any unmeasured variables shown in a factor will be excluded in the factor's embedding factor only when the embedding factor is obtained by marginalizing out those unmeasured variables.

We provide another example for obtaining an MVEF which requires multiple steps using the graph in Fig. 9a. Given $P_E(D, X, Y_2)$ co-identifying $P_{b,e}(y_2)$ takes two steps. First, B is marginalized out leading to its embedding factor,

$$\sum_b P(a,b) P_{b,e}(y_2) = P_e(a, y_2).$$

Then, again A is taken out yielding,

$$\sum_a P_{a,x}(d) P_e(a, y_2) = P_{e,x}(d, y_2).$$

Proposition 4 (Uniqueness)**.** *If an MVEF exists for a gc-factor with respect to a distribution, then it is unique.*

Proof. Consider an MVEF for a gc-factor in $\mathbb{F}$ with $P_{\mathbf{Z}'}(\mathbf{V}')$. An MVEF must satisfy that the $\mathbf{V}'$ and $\mathbf{Z}'$ cover the embedding factor (the two inclusion criteria). Only way to do it is marginalizing out those uncovered since otherwise

Algorithm 2 MVEF and MVEF-admissible set

Input: $\mathcal{G}$ a causal graph, $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$ a gc-factor in $\mathbb{F}$, $\mathbf{Z}'$ and $\mathbf{V}'$ representing $P_{\mathbf{Z}'}(\mathbf{V}')$
$\mathbf{V}'' \leftarrow \mathbf{V}^+$
while TRUE **do**
  let $\langle \mathbf{X}'', \mathbf{Y}'' \rangle \in \mathbb{F}''$ be the embedding factor of $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$
  if $\mathbf{Y}'' \cap \mathbf{V}^* \not\subseteq \mathbf{V}'$ **or** $\mathbf{X}'' \cap \mathbf{V}^* \not\subseteq \mathbf{V}' \cup \mathbf{Z}'$ **or** $\mathbf{Z}' \cap An(\mathbf{Y}'')_{\mathcal{G}_{\underline{\mathbf{X}''}}} \neq \emptyset$ **then**
    return NULL
  end if
  if $\mathbf{Y}'' \subseteq \mathbf{V}'$ and $\mathbf{X}'' \subseteq \mathbf{V}' \cup \mathbf{Z}'$ **then**
    return $\langle \mathbf{X}'', \mathbf{Y}'' \rangle$, $\mathbf{V}^+ \setminus \mathbf{V}''$
  end if
  $\mathbf{V}'' \leftarrow \mathbf{V}'' \setminus ((\mathbf{Y}'' \setminus \mathbf{V}') \cup (\mathbf{X}'' \setminus (\mathbf{V}' \cup \mathbf{Z}')))$
end while

they will be left uncovered (Prop. 3). This implies that one can iteratively obtain projections till the inclusion criteria are satisfied. If the criteria are satisfied, then the undisturbed mechanism criterion is examined. Moreover, by Prop. 11 the set $An(\mathbf{Y}^\dagger)_{\mathcal{G}_{\underline{\mathbf{X}^\dagger}}}$ only grows along the embedding order, so a disturbed mechanism at the minimal inclusion-satisfying level is disturbed at every coarser level and no further projection can restore the third criterion. If all three criteria hold true, then the embedding factor is a unique MVEF. □

Proposition 5 (Disjointness)**.** *Given a distribution, let Q and R be MVEFs of two gc-factors in $\mathbb{F}$, respectively. Then either $Q = R$ or there is no common gc-factor embedded in Q and R.*

Proof. Factors are merged with respect to the lack of variables in $\mathbf{Y}^+$ (i.e., $\mathbf{Y}^+ \setminus \mathbf{V}'$). By the minimality and Prop. 3, if they are not disjoint, they should be merged into the same factors. □

Following propositions examine graphical properties of a factor and its embedding factor. These propositions will be crucial in proving Lemma 4 (*hedge over embedding factors*).

Proposition 10. *Let $\langle \mathbf{X}'', \mathbf{Y}'' \rangle \in \mathbb{F}''$ be a proper embedding factor of $\langle \mathbf{X}', \mathbf{Y}' \rangle \in \mathbb{F}'$ where $\mathbf{V}' \setminus \mathbf{V}'' = \{W\}$ and $W \in (\mathbf{X}' \cup \mathbf{Y}') \setminus \mathbf{V}^*$. Then, $An(\mathbf{Y}')_{\mathcal{G}_{\underline{\mathbf{X}'}}} \subseteq An(\mathbf{Y}'')_{\mathcal{G}_{\underline{\mathbf{X}''}}}$.*

Proof. If $W \in \mathbf{Y}'$, then $\mathbf{Y}''$ includes $\mathbf{Y}' \setminus \{W\}$ and $ch(W)_{\mathcal{G}'} \cap \mathbf{Y}^{+\prime}$, which is non-empty because $W \in \mathbf{Y}^{+\prime} \setminus \mathbf{V}^*$. Thus, $An(\mathbf{Y}')_{\mathcal{G}} \subseteq An(\mathbf{Y}'')_{\mathcal{G}}$. Further, $\mathbf{X}''$ consists of $\mathbf{X}' \setminus \mathbf{Y}''$ and the ancestors of $\mathbf{X}' \cap \mathbf{Y}''$ without intersecting $An(\mathbf{Y}')_{\mathcal{G}_{\underline{\mathbf{X}'}}}$. Hence, $An(\mathbf{Y}')_{\mathcal{G}_{\underline{\mathbf{X}'}}} \subseteq An(\mathbf{Y}'')_{\mathcal{G}_{\underline{\mathbf{X}''}}}$.

If $W \in \mathbf{X}'$, then $\mathbf{Y}''$ contains $\mathbf{Y}'$. $\mathbf{X}''$ includes $\mathbf{X}'$ excluding W and those intersecting with $\mathbf{Y}''$. However, $\mathbf{X}''$ contains the ancestors of $\mathbf{Y}'' \cup \{W\}$, that is, any variables removed from $\mathbf{X}'$ will be replaced by its ancestors. Further, $\mathbf{X}''$ is disjoint with $An(\mathbf{Y}')_{\mathcal{G}_{\underline{\mathbf{X}'}}}$. Therefore, $An(\mathbf{Y}')_{\mathcal{G}_{\underline{\mathbf{X}'}}} \subseteq An(\mathbf{Y}'')_{\mathcal{G}_{\underline{\mathbf{X}''}}}$. □

Proposition 11. *Let $\langle \mathbf{X}'', \mathbf{Y}'' \rangle \in \mathbb{F}''$ be an embedding factor of $\langle \mathbf{X}', \mathbf{Y}' \rangle \in \mathbb{F}'$. Then, $An(\mathbf{Y}')_{\mathcal{G}_{\underline{\mathbf{X}'}}} \subseteq An(\mathbf{Y}'')_{\mathcal{G}_{\underline{\mathbf{X}''}}}$.*

Proof. One can marginalize out $\mathbf{W} \subseteq \mathbf{V}' \setminus \mathbf{V}''$ in an order, so as to yield $\langle \mathbf{X}'', \mathbf{Y}'' \rangle$ (Prop. 3). It follows from the transitivity of subset relationships and Prop. 10. □

Proposition 12. *An MVEF-admissible set corresponds to $an(\mathbf{Y}^\dagger)_{\mathcal{G}_{\underline{\mathbf{X}^\dagger}}} \setminus \mathbf{Y}^\dagger$, the variables lied between $\mathbf{Y}^\dagger$ and $\mathbf{X}^\dagger$.*

Proof. This follows from the minimality and Prop. 3 that only variables changing the embedding factor will be projected out. □

Proposition 13. *Let $\langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle \in \mathbb{F}^\dagger$ be the MVEF of $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle \in \mathbb{F}$ with respect to $P_{\mathbf{Z}'}(\mathbf{V}')$. Any factor in $\mathbb{F}''$ embedding in $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$ does not satisfy the three criteria if $\mathbf{V}'' \not\subseteq \mathbf{V}^\dagger$.*

Proof. By Prop. 3, the variables in $\mathbf{V}'' \setminus \mathbf{V}^\dagger$ will not be covered by the inclusion criteria. □

Lemma 4 (Hedge over Embedding Factors)**.** *Consider a gc-factor $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle \in \mathbb{F}$ and a distribution $P_{\mathbf{Z}'}(\mathbf{V}')$. The gc-factor is not co-identifiable with $P_{\mathbf{Z}'}(\mathbf{V}')$ with respect to any of its embedding factors if its MVEF does not exist or there exists a hedge for $P_{\mathbf{x}^\dagger \setminus \mathbf{Z}'}(\mathbf{y}^\dagger)$ in $(\mathcal{G} \setminus \mathbf{Z}')\langle \mathbf{V}' \rangle$ given the MVEF $\langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle \in \mathbb{F}^\dagger$.*

Proof. By definition, if the MVEF of $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$ does not exist, then no embedding factor of $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$ satisfies the first three criteria simultaneously. Hence, $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$ is not co-identifiable with $P_{\mathbf{Z}'}(\mathbf{V}')$ (Cor. 1). We now focus only on the factors that embed the MVEF following Prop. 13. Even among the embedding factors of the MVEF, we can rule out those failed to meet the three criteria. Therefore, we focus on the embedding factors of the MVEF satisfying the three criteria to show that such factors also incur a hedge and, thus, non-identifiable.

Let $\langle \mathcal{F}, \mathcal{F}' \rangle$ be a hedge rooted on $\mathbf{R}$ for $P_{\mathbf{x}^\dagger \setminus \mathbf{Z}'}(\mathbf{y}^\dagger)$ in $\mathcal{G}' = (\mathcal{G} \setminus \mathbf{Z}')\langle \mathbf{V}' \rangle$ (Cor. 1).[7] Let $\mathbf{T}^\dagger = \mathbf{X}^\dagger \setminus \mathbf{Z}'$. We first characterize the hedge.

(Claim 1) $\mathbf{R} \subseteq \mathbf{Y}^\dagger$. The parents of $\mathbf{Y}^\dagger$ in $\mathcal{G}'$ are $\mathbf{T}^\dagger$ since the parents of $\mathbf{Y}^\dagger$ are $\mathbf{X}^\dagger$ in $\mathcal{G}^\dagger$ and $\mathcal{F}' \cap \mathbf{T}^\dagger$ must be empty by definition.

(Claim 2) There exists a hedge with $\mathbf{R} = \mathbf{Y}^\dagger$. On top of Claim 1, attaching $\mathbf{Y}^\dagger \setminus \mathbf{R}$ to $\mathcal{F}'$ does not break the hedgeness of $\langle \mathcal{F}, \mathcal{F}' \rangle$ since $\mathbf{Y}^\dagger$ are all confounded in $\mathcal{G}'$. Without loss of generality, we let $\mathbf{R} = \mathbf{Y}^\dagger$ to reduce an additional layer of complexity.

[7] For the sake of readability, this does not follow our convention that $\mathcal{G}' = \mathcal{G}\langle \mathbf{V}' \rangle$.

(Claim 3) $\mathbf{T}^\dagger \cap \mathcal{F} \neq \emptyset$. This is immediate from the definition of a hedge formed for $P_{\mathbf{t}^\dagger}(\mathbf{y}^\dagger)$. It need *not* hold that $\mathbf{X}^* \cap \mathcal{F} \neq \emptyset$: nothing forces $\mathbf{X}^* \subseteq \mathbf{V}'$, and after projecting $\mathbf{V}^\dagger \setminus \mathbf{V}'$ out, $\mathbf{Y}^\dagger$ can be confounded with variables of $\mathbf{Y}^{+\prime} \setminus \mathbf{Y}^\dagger$ rather than with $\mathbf{X}^+ \cap \mathbf{V}'$ alone.

Consider now an arbitrary factor $\langle \mathbf{X}'', \mathbf{Y}'' \rangle \in \mathbb{F}''$ which embeds the MVEF and satisfies the three criteria. We examine the existence of a hedge for $P_{\mathbf{t}''}(\mathbf{y}'')$ in $\mathcal{G}'$ where $\mathbf{T}'' = \mathbf{X}'' \setminus \mathbf{Z}'$.

(Claim 4) $\mathbf{T}'' \cap \mathcal{F} \neq \emptyset$. This is what the rest of the proof uses, and it follows from Prop. 11 with the preservation of ancestral relationships under projection and the definition of gc-factor where $\mathbf{Y}$-side is in $\mathbf{Y}^{+\prime}$ and $\mathbf{X}$-side cannot intersect $\mathbf{X}^+ \setminus \mathbf{X}^*$. The stronger "betweenness" form — that $(\mathbf{T}'' \cap \mathcal{F})$, $De(\mathbf{X}^* \cap \mathcal{F})_{\mathcal{F}}$ and $An(\mathbf{T}^\dagger \cap \mathcal{F})_{\mathcal{F}}$ have a common element — does not hold in general, with either $\mathbf{X}^*$ or $\mathbf{T}^\dagger$ in the $De(\cdot)$ slot, and is not needed.

(Claim 5) $\mathbf{T}'' \cap \mathcal{F}' = \emptyset$. Let $\mathbf{W}$ be those variables further projected out relative to the MVEF, i.e., $\mathbf{V}^\dagger \setminus \mathbf{V}''$. It is by definition $\mathbf{W} \cap \mathbf{X}'' = \emptyset$ and, thus, $\mathbf{W} \cap \mathbf{T}'' = \emptyset$. Since $\mathbf{V}_{\mathcal{F}'} = \mathbf{R} = \mathbf{Y}^\dagger \subseteq \mathbf{Y}'' \cup \mathbf{W}$ (Claim 2) and $\mathbf{T}'' \cap \mathbf{Y}'' = \emptyset$ holds by definition of gc-factor, $\mathbf{T}'' \cap \mathcal{F}' = \emptyset$ follows.

Finally, $\mathbf{R} \subseteq An(\mathbf{Y}^\dagger)_{\mathcal{G}'_{\underline{\mathbf{T}^\dagger}}} \subseteq An(\mathbf{Y}'')_{\mathcal{G}'_{\underline{\mathbf{T}''}}}$ holds by Prop. 11 and the conservation of ancestral relationships under projections. Further, $\mathbf{T}'' \cap \mathcal{F} \neq \emptyset$ (Claim 4) and $\mathbf{T}'' \cap \mathcal{F}' = \emptyset$ (Claim 5). Hence, $\langle \mathcal{F}, \mathcal{F}' \rangle$ is also a valid hedge for $P_{\mathbf{x}''}(\mathbf{y}'')$ with respect to $P_{\mathbf{Z}'}(\mathbf{V}')$ in $\mathcal{G}$, that is, for $P_{\mathbf{t}''}(\mathbf{y}'')$ in $\mathcal{G}'$. Consequently, any embedding factor of the MVEF is not identifiable with $P_{\mathbf{Z}'}(\mathbf{V}')$. □

Lemma 5 (Compositionality)**.** *Given a distribution $P_{\mathbf{Z}'}(\mathbf{V}')$, any identifiable embedding factor of a gc-factor in $\mathbb{F}$ can be represented as the summation over the product of a subset of identified MVEFs.*

Proof. An embedding factor can be characterized by its embedded factors and the variables marginalized out. First, by the definition of MVEF and hedge over embedding factors (Lemma 4), no factor that embeds any non-co-identified gc-factor is identifiable. Hence, only co-identified gc-factors can be included in an identifiable embedding factor.

If any co-identified gc-factor embedded in the embedding factor is not embedded together with other gc-factors embedded in its MVEF, then it implies that an unmeasured variable is remained in the embedding factor.[8] Hence, an embedding factor is identifiable only if it is composed of the identified MVEFs. □

[8] For example, consider a distribution $P_X(A, C) \in \mathbb{P}$ and a factor $P_x(b) \in \mathbb{F}'$ with $\mathbf{V}' = \{B, C, X, Y\}$ in Fig. 7a. $P_x(b)$ (i.e., the embedding factor) embeds $P(a)$ and $P_{ax}(b)$ (i.e., a co-identified gc-factor) but not $P_b(c)$ (i.e., other gc-factors embedded in its MVEF). $P_x(b)$ is not identifiable from $P_X(A, C)$ as it still includes B (i.e., an unmeasured variable is remained in the embedding factor).

Theorem 1 (Soundness)**.** GID-PO *is sound.*

Proof. The correctness of identified MVEF from one of available distribution is provided in Cor. 1, the general single-distribution result; Thm. 2 covers only the observational case $\mathbf{Z}' = \emptyset$. Further, let $\mathbf{i}_j$ be the indices of the gc-factors embedded in the j-th selected MVEF and $\mathbf{W}_j$ its MVEF-admissible set. The exact cover makes $\{\mathbf{i}_j\}_j$ a partition of the indices of $\mathbb{F}$. By the definition of an embedding factor and Prop. 3, every gc-factor in $\mathbb{F}$ containing a variable of $\mathbf{W}_j$ belongs to $\mathbf{i}_j$, so marginalizing $\mathbf{W}' = \bigcup_j \mathbf{W}_j$ merges precisely the gc-factors within each $\mathbf{i}_j$ and never merges two selected blocks; the gc-factors of $\mathbb{F}^{\mathbf{V}^+ \setminus \mathbf{W}'}$ are therefore exactly the selected MVEFs, and Prop. 2 yields the returned sum-product formula with the outer summation over $\mathbf{Y}^+ \setminus (\mathbf{Y} \cup \mathbf{W}')$. □

E. Discussion on the NP-completeness of GID-PO

Proposition 14. *Let $P_{\mathbf{Z}'}(\mathbf{V}')$ and $P_{\mathbf{Z}''}(\mathbf{V}'')$ be two distributions corresponding to two different subsets in $2^{[n]} \setminus \{\emptyset\}$. $P_{\mathbf{Z}'}(\mathbf{V}')$ is not identifiable from $P_{\mathbf{Z}''}(\mathbf{V}'')$ in $\mathcal{G}$.*

Proof. Let $\mathbf{W}'$ and $\mathbf{W}''$ be the two different nonempty subsets of $[n]$. If one set is not subsumed by other set, there exists an integer i which appears only in one set and j in other set such that $i < j$. By the construction, $V_{i,j}$ and $V_{j,i}$ are measured in the distributions corresponding to sets having i and j, respectively. Hence, outcome inclusion criterion is failed in either direction. For example, consider $\{1, 2\}$ and $\{1, 3\}$. One has $V_{2,3}$ and the other has $V_{3,2}$ as outcome.

Otherwise, one of the two sets is strictly contained in the other. Let first $\mathbf{W}' \subsetneq \mathbf{W}''$ (a larger set implies that there are more variables projected out to combine gc-factors). There must be a variable projected out in the distribution for $\mathbf{W}''$ but not in the one for $\mathbf{W}'$: pick $i \in \mathbf{W}'$ and $k \in \mathbf{W}'' \setminus \mathbf{W}'$ and take $V_{i,k}$ if $i < k$ and $V_{k,i}$ otherwise; such a choice exists because $\mathbf{W}' \neq \emptyset$. The distribution for $\mathbf{W}''$ cannot identify that for $\mathbf{W}'$. If the variable is the part of $\mathbf{V}'$, then it does not satisfy the outcome inclusion criterion. If the variable is the part of $\mathbf{Z}'$, then it fails to meet the minimal treatment inclusion criterion. For example, consider $\{1, 2, 3\}$ versus $\{1, 2\}$ and versus $\{1, 3\}$. $V_{2,3}$ in the outcome and $V_{2,3}$ in the treatment are missed, respectively. Let finally $\mathbf{W}'' \subsetneq \mathbf{W}'$, the reverse case, which is not covered by the above since identifiability is directional. Pick $i \in \mathbf{W}' \setminus \mathbf{W}''$ and $j \in \mathbf{W}''$. If $i > j$, then $V_{i,j}$ is an outcome of the $\mathbf{W}'$-distribution and is absent from the $\mathbf{W}''$-distribution, so the outcome inclusion criterion fails. If $i < j$, then $V_{i,j}$ is intervened on in the

Algorithm 3 GTR-PO

Input: $\mathcal{G}$ a causal graph, $\mathbf{x}$ and $\mathbf{y}$ value assignments for a query, $\{\mathbb{P}^k\}_{k=1}^q$ a collection of available distributions from q heterogeneous domains, $\mathbf{\Delta} = \{\Delta_k\}_{k=1}^q$ a collection of sets of variables pointed by selection variables for each domain. (We write q for the number of domains, reserving $m = |\mathbb{P}|$ for the number of available distributions.)
Prepare $\mathbb{J}$ an empty collection.
for $k \in [q]$ **do**
  for $\langle P^k_{\mathbf{Z}'}(\mathbf{V}'), \langle \mathbf{X}_i, \mathbf{Y}_i \rangle \rangle \in \mathbb{P}^k \times \mathbb{F}$ **do**
    continue if an MVEF that embeds $\langle \mathbf{X}_i, \mathbf{Y}_i \rangle$ is already found by the same data $P^k_{\mathbf{Z}'}(\mathbf{V}')$
    $\langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle, \mathbf{W}^\dagger \leftarrow \text{MVEF}(\langle \mathbf{X}_i, \mathbf{Y}_i \rangle, P^k_{\mathbf{Z}'}(\mathbf{V}'))$.
    continue if the same MVEF $\langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle$ is already recorded in $\mathbb{J}$ from some data.
    if $\langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle$ exists **and** identifiable with $P^k_{\mathbf{Z}'}(\mathbf{V}')$ **and** $An(\mathbf{Y}^\dagger)_{\mathcal{G}_{\underline{\mathbf{X}^\dagger}}} \cap \Delta_k = \emptyset$ **then**
      $\mathbf{i} \leftarrow$ the indices of factors in $\mathbb{F}$ embedded in the MVEF.
      Add $\langle \langle \mathbf{X}^\dagger, \mathbf{Y}^\dagger \rangle, \mathbf{W}^\dagger, \mathbf{i}, \langle \mathbf{Z}', \mathbf{V}' \rangle, k \rangle$ to $\mathbb{J}$.
    end if
  end for
end for
Let $\mathbb{I}$ be the collection of indices in $\mathbb{J}$.
if $\mathbb{I}' \leftarrow$exact cover with $\{1, \ldots, |\mathbb{F}|\}$ and $\mathbb{I}$ **then**
  Let $\mathbb{J}' \subseteq \mathbb{J}$ be the subcollection matching the solution $\mathbb{I}'$.
  Let $\mathbf{W}'$ be the union of all MVEF-admissible sets in $\mathbb{J}'$.
  return $\sum_{\mathbf{y}^+ \setminus (\mathbf{Y} \cup \mathbf{W}')} \prod_{\mathbb{J}'} \text{ID}(P^k_{\mathbf{x}^\dagger \setminus \mathbf{Z}'}(\mathbf{y}^\dagger), (\mathcal{G} \setminus \mathbf{Z}')\langle \mathbf{V}' \cap \mathbf{V}^+ \rangle, P^k_{\mathbf{z}'})$.
end if
return NULL.

$\mathbf{W}''$-distribution and is a directed ancestor of the outcome $V_{j,i}$ of the $\mathbf{W}'$-query without being one of its treatments, so $\mathbf{Z}'' \cap \mathbf{Y}^+ \neq \emptyset$ and the undisturbed-mechanism criterion fails; both inclusion criteria may well hold in this case (they do for $\mathbf{W}' = [k]$ and $\mathbf{W}'' = [k] \setminus \{1\}$). □

F. Discussion for Transportability and Conditional Interventional Distribution

Extension to Transportability The key question in transportability (Pearl & Bareinboim, 2011; Bareinboim & Pearl, 2014; Lee et al., 2020) is whether factors that constitute the given query can be compatible between the target and a source domain. This can be tested by whether the differences between the two domains represented as *selection variables* $\mathbf{S}$, auxiliary variables attached to the given causal diagram resulting in a *selection diagram* $\mathcal{H}$, can be ignorable in estimating the probability of interest. Graphically speaking, this coincides with whether $\mathbf{Y}^\dagger$, the $\mathbf{Y}$ side of a factor, is independent to the set of selection variables in the post-intervention graph $\mathcal{H}_{\overline{\mathbf{X}^\dagger}}$. Formally,

$$(\mathbf{S}^k \perp\!\!\!\perp \mathbf{Y}^\dagger \mid \mathbf{X}^\dagger)_{\mathcal{H}_{\overline{\mathbf{X}^\dagger}}}. \tag{6}$$

Algorithm 4 GTRC-PO

Input: $\mathbf{w}$ a value assignment for a query $P_{\mathbf{x}}(\mathbf{y}|\mathbf{w})$ in addition to inputs for GTR-PO.
if exists $W \in \mathbf{W}$ s.t. $(W \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W} \setminus \{W\})_{(\mathcal{G}\setminus\mathbf{X})_{\underline{W}}}$ **then**
  return GTRC-PO$(\mathbf{y}, \mathbf{x} \cup \{w\}, \mathbf{w} \setminus \{w\}, \mathcal{G}, \{\mathbb{P}^k\}_{k=1}^q, \mathbf{\Delta})$.
else
  return $\frac{Q(\mathbf{y},\mathbf{w})}{\sum_{\mathbf{y}' \in \mathfrak{X}_\mathbf{Y}} Q(\mathbf{y}',\mathbf{w})}$ where $Q(\tilde{\mathbf{y}}, \mathbf{w}) =$ GTR-PO$(\tilde{\mathbf{y}} \cup \mathbf{w}, \mathbf{x}, \mathcal{G}, \{\mathbb{P}^k\}_{k=1}^q, \mathbf{\Delta})$.
end if

which is identical to $\mathbf{Y}^\dagger \cap \Delta_k = \emptyset$ under full-observability. However, in the partial-observability setting, there are variables projected out in-between $\mathbf{Y}^\dagger$ and $\mathbf{X}^\dagger$ (Prop. 12). Hence, an equality test corresponding to the conditional independence (Eq. (6)) is

$$An(\mathbf{Y}^\dagger)_{\mathcal{G}_{\underline{\mathbf{X}^\dagger}}} \cap \Delta_k = \emptyset. \tag{7}$$

We present the resulting transportability algorithm which takes partial-observability into account in Alg. 3. The only difference is the use of k to denote which domain a distribution comes from and an additional condition at Line 8 (Eq. (7)).

Extension to a Conditional Query A conditional interventional probability $P_{\mathbf{x}}(\mathbf{y}|\mathbf{w})$ can be computed using marginal probabilities as

$$P_{\mathbf{x}}(\mathbf{y} \mid \mathbf{w}) = \frac{P_{\mathbf{x}}(\mathbf{y}, \mathbf{w})}{P_{\mathbf{x}}(\mathbf{w})} = \frac{P_{\mathbf{x}}(\mathbf{y}, \mathbf{w})}{\sum_{\mathbf{y}' \in \mathfrak{X}_\mathbf{Y}} P_{\mathbf{x}}(\mathbf{y}', \mathbf{w})}.$$

based on the definition of conditional probability. Therefore, the task can be easily delegated to our main problem: general identifiability (or transportability) under partial-observability.

However, since not every $\mathbf{W}$ can be relevant to $\mathbf{Y}$, some of them can be either completely ignorable or can be exchanged to intervention. Shpitser & Pearl (2006b) describes such procedure which utilizes Rule 2 of do-calculus to maximally delegate observations to interventions. As a consequence, $\mathbf{Y}$ and the remaining $\mathbf{W}$, say $\mathbf{W}'$, in observation can be characterized by the existence of a back-door path between $Y \in \mathbf{Y}$ and $W \in \mathbf{W}'$. It is now well-understood that $\mathbf{W}'$ are essential in estimating $P_{\mathbf{x},\mathbf{w}\setminus\mathbf{w}'}(\mathbf{y}|\mathbf{w}') = P_{\mathbf{x}}(\mathbf{y}|\mathbf{w})$ under full-observability (Lee et al., 2020).

This procedure, which only involves in a given query, is sound. The question of the completeness of identifying a conditional causal effect under the assumption of the completeness of unconditional causal effect needs to be carefully answered but we conjecture that if GID-PO is complete, then the aforementioned procedure will be complete, too. The procedure is shown in Alg. 4, which calls for Alg. 3.

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Changes

Unofficial AI-corrected version · 2026-10-25 · Author review pending

Original publication. Sanghack Lee and Elias Bareinboim. *Causal Effect Identifiability under Partial-Observability*. Proceedings of the 37th International Conference on Machine Learning, PMLR 119:5692–5701, 2020. Proceedings record. Baseline: Technical Report R-58 (causalai.net/r58.pdf, last revision November 2020), whose main text is identical to the PMLR paper and which includes Appendices A–F. Prepared with AI assistance; these proposed corrections are unofficial and are not a publisher-issued correction or author approval. The numbering of sections, definitions, results, figures and algorithms is unchanged; equation numbers refer to this version, in which Eq. (5) (proof of Lemma 7) is new. Appendix items apply to the technical report.

Location	Before → After	Assumptions / limits
Def. 1, footnote 1; §2	Divergent path: directed paths $(X, \dots, W_1)$ and $(W_2, \dots, Y)$ with $W_1 = W_2$ or $W_1 \leftrightarrow W_2 \rightarrow$ two vertex-disjoint directed paths towards X and Y (each possibly trivial) joined by $W_1 \leftrightarrow W_2$, or a common ancestor $W \notin \{X, Y\}$; equivalently, all internal vertices are non-colliders and both ends carry arrowheads. $\mathcal{G} \setminus \mathbf{W}$ is defined as vertex deletion, $\mathcal{G}[\mathbf{V} \setminus \mathbf{W}]$, distinct from the projection $\mathcal{G}\langle \text{-}\mathbf{W}\rangle$.	Coincides with the standard latent projection (checked on all ADMGs with at most four vertices). Read literally, the printed footnote adds $X \leftrightarrow Y$ beside every directed edge.
Lemma 1; Cor. 1; Lemma 2	$\mathbf{z}'$ "consistent with $\mathbf{x}^+ \cap \mathbf{Z}'$" ($\mathbf{x}^+$ undefined) → any $\mathbf{z}'$ agreeing with $\mathbf{x}$ on $\mathbf{X}^* \cap \mathbf{Z}'$. Lemma 2: $P^1_{\mathbf{Z}_i}(\mathbf{V} \setminus \mathbf{Z}_i) = P^2_{\mathbf{Z}_i}(\mathbf{V} \setminus \mathbf{Z}_i) > 0 \rightarrow$ each model positive.	Conclusions unchanged; Lemma 2 assumes only what Def. 4 and its proof use.
Def. 8; Alg. 2 (MVEF), lines 2, 5, 8, 9, 11	$\mathbf{X}''$ and $\mathbf{Y}''$ exchanged relative to Def. 8 in lines 5, 8, 11, and NULL returned for an *undisturbed* mechanism ($= \emptyset$) → outcome side checked against $\mathbf{V}'$, treatment side against $\mathbf{V}' \cup \mathbf{Z}'$, and the mechanism-failure disjunct becomes $\mathbf{Z}' \cap An(\mathbf{Y}'')_{\mathcal{G}_{\overline{\mathbf{X}''}}} \neq \emptyset$. $\mathbf{V}'' \leftarrow \mathbf{V}$ and return $\mathbf{V} \setminus \mathbf{V}'' \rightarrow \mathbf{V}^+$ and $\mathbf{V}^+ \setminus \mathbf{V}''$; Def. 8's admissible set $\mathbf{V} \setminus \mathbf{V}^\dagger \rightarrow \mathbf{V}^+ \setminus \mathbf{V}^\dagger$.	Def. 8 and its example already use this orientation (brute-force enumeration agrees); the returned admissible set now lies within $\mathbf{Y}^+$ (Prop. 12).
Alg. 1, lines 6, 16; Alg. 3, lines 7, 18; Alg. 4	GID$(P_{\mathbf{x}^\dagger}(\mathbf{y}^\dagger), \mathcal{G}\langle \mathbf{V}'\rangle, \{\mathbf{Z}'\})$ with $\mathbf{Z}'$ outside $\mathcal{G}\langle \mathbf{V}'\rangle$, index $\mathbf{y}^{+\prime} \rightarrow$ Cor. 1's form ID$(P_{\mathbf{x}^\dagger \setminus \mathbf{Z}'}(\mathbf{y}^\dagger), (\mathcal{G} \setminus \mathbf{Z}')\langle \mathbf{V}' \cap \mathbf{V}^+\rangle, P_{\mathbf{z}'})$, index $\mathbf{y}^+$. "continue if the MVEF is co-identified by some data" → continue if the same MVEF is already in $\mathbb{J}$. Alg. 4: recursive calls carry $\{\mathbb{P}^k\}_{k=1}^q$; $Q(\tilde{\mathbf{y}}, \mathbf{w})$ is a function.	Line 6 (Alg. 3, line 7) is read as de-duplication; it never removes a candidate from $\mathbb{I}$.
§4.2, complexity; §5, reduction	"…MVEFs, which is upper bounded by $\lvert\mathcal{F}\rvert$"; $O(\ell m n^4 + 2^\ell) \rightarrow$ at most ℓm identified MVEFs, representing at most $\min(\ell m,\ 2^\ell - 1)$ distinct index sets of embedded factors; subset-DP exact cover in $O(2^\ell \lvert\mathbb{J}\rvert)$; total $O(\ell m \lvert\mathbf{V}\rvert^4 + \ell m 2^\ell)$. Reduction: $\mathbf{V} = \{V_{i,j}\}_{i<j}$ and $\mathbf{V}^\dagger = \mathbf{V} \setminus \{V_{i,j}\}_{i \neq j;\ i,j \in \mathbf{w}} \rightarrow i \neq j$ and $i < j$, respectively; the bound is stated in $\lvert\mathbf{V}\rvert$ and §5's universe is $[k]$.	"Exponential only in $\lvert\mathcal{F}\rvert$" and polynomial time under full observability are unchanged; the paper's own reduction gives seven MVEFs for $\lvert\mathcal{F}\rvert = 3$.
§5, "On Completeness", "On NP-Completeness"	Two-model recipe (fair coins, exclusive-or, one flipped Z') offered as formal support → a positive spanning-tree parity construction in which each treatment enters exactly once (unused parent arguments ignored), with its closed-form kernel. Reduction from exact cover (NP-hardness only) → NP-membership certificate added: a level $\mathbf{V}'$, $\mathbf{V}^* \subseteq \mathbf{V}' \subseteq \mathbf{V}^+$, with an identifying distribution per gc-factor, checked by Cor. 1.	The completeness conjecture is left as printed (note below); NP-completeness remains conditional on it. The printed recipe is separated by $P(\mathbf{Z})$, a smaller piece, already for $W \rightarrow Z_1, W \leftrightarrow Z_2, Z_1 \leftrightarrow Z_2$.
Lemma 7 (Hedge Exclusion), proof; Examples 1–2; Fig. 13 (App. B–C)	$P^1(\mathbf{V}_\mathcal{H}) = P^2(\mathbf{V}_\mathcal{H}) > 0$, per-path bits with a root-noise count m, limits "close to 0.5" and "close to 1" for $r = 0$, noiseless exclusive-or, $\varepsilon \neq 0.5 \rightarrow$ the proof is rewritten with linear models: one coordinate per realized path, an independent private noise or private fair coin for each coordinate, the vertices of $\mathcal{F}'$ in $\mathcal{M}^2$ retaining only their $\mathcal{F}'$ inputs, one compensating noise with $1 - 2\delta = \prod_{N \in J^1 \setminus J^2}(1 - 2\varepsilon_N)$ over the noises $\mathcal{M}^2$ drops, and the root parity with $P^1_{\mathbf{x}^*}(\bigoplus \mathbf{R}{=}0) = \frac{1}{2}$ versus $\frac{1}{2}(1 + \gamma)$. Examples 1–2: $P^2(U_Y{=}1) = \varepsilon + \delta - 2\varepsilon\delta$, $0 < \varepsilon, \delta < 0.5$; Fig. 13: $1 - 2\delta = (1 - 2\varepsilon)^6$.	The printed full-joint claim is impossible for Fig. 12(b); the $r = 0$ limit holds only for $\lvert\mathbf{R}\rvert = 1$; the statements of Lemma 7 and Thm. 2 are unchanged. Exact tests passed on the two earlier counterexamples, Fig. 13 and 500 random hedges with admissible random liftings.
Props. 10, 13; proofs of Thm. 1, Lemma 4, Props. 4, 7, 8, 14 (App. B, D–E)	Prop. 10: $W \in (\mathbf{X}'' \cup \mathbf{Y}'') \setminus \mathbf{V}^* \rightarrow W \in (\mathbf{X}' \cup \mathbf{Y}') \setminus \mathbf{V}^*$. Prop. 13: $\mathbf{V}^\dagger \not\subseteq \mathbf{V}'' \rightarrow \mathbf{V}'' \not\subseteq \mathbf{V}^\dagger$. Lemma 4, Claims 3–5: $\mathbf{X}^* \cap \mathcal{F} \neq \emptyset$ and a betweenness step → $\mathbf{T}^\dagger \cap \mathcal{F} \neq \emptyset$ and $\mathbf{T}'' \cap \mathcal{F} \neq \emptyset$ only. Prop. 14: both strict-containment cases covered, by outcome inclusion when $i > j$ and by mechanism disturbance when $i < j$. Prop. 4: Prop. 11 invoked. Prop. 7: U_Y exempted from sharing. Prop. 8: off-path mechanisms ignore X. Thm. 1: Cor. 1 identifies the selected MVEFs; exact-cover disjointness with Props. 2–3 justifies their assembly.	Only Props. 10 and 13 change their statements. Finite checks support the algorithmic and graphical repairs but are not proofs; re-basing Lemma 4's betweenness on $\mathbf{T}^\dagger$ is also false.
Throughout	Example after Fig. 6: U_A noise restores positivity ($0.9 \rightarrow 0.82$). Remark after Def. 6 qualified ($\mathbf{Y}$ a single c-component of $\mathcal{G}\langle \mathbf{V}^*\rangle$). Baseline $\mathcal{F}_{\mathcal{G},\mathbf{V}^+}$. §5's transportability sentence broadened to App. F, Eq. (7), including projected-out ancestral mechanisms. Fig. 3(b) vertices and Fig. 7's caption footnote restored; Fig. 9 caption; Fig. 10(b) outcome region completed, a stray spacer label hidden. Notation ($\mathbf{Y}^{+\prime}$, $\mathcal{G}[\mathbf{V} \setminus \{W\}]$, $\mathbf{S}^k$, q, Δ_k, $P_{\mathbf{x},\mathbf{w} \setminus \mathbf{w}'}(\mathbf{y} \mid \mathbf{w}')$), the MVEF example's projection criteria, model labels, Remark 1 items in Prop. 9's proof, a cross-reference, grammar.	The example after Fig. 6 keeps its conclusion; the remark after Def. 6 now carries its condition; the transportability sentence restates Eq. (7); the rest are local proof, notation and presentation fixes.

Completeness conjecture. §5 and §6 conjecture that Lemma 3 is also necessary (that GID-PO is complete) and condition the NP-completeness discussion on it. The conjecture is left as printed; it was later shown to be false by the authors (S. Lee and E. Bareinboim, *Causal identification with matrix equations*, Advances in Neural Information Processing Systems 34, 2021, Prop. 1): for $X \rightarrow A \rightarrow B \rightarrow Y$ with finite discrete variables and $P(X, B)$, $P(A, B)$ and $P(A, Y)$, $P_x(y) = P(y \mid A)\, P(B \mid A)^\dagger P(B \mid x)$ holds in every compatible model (†: Moore–Penrose pseudoinverse), while no level $\mathbf{V}'$ satisfies Lemma 3 and GID-PO returns NULL. Lemma 3 and Thm. 1 (soundness) do not depend on the conjecture.

Verification scope. The paper has no empirical experiments. Exact arithmetic confirms the corrected Examples 1–2, the example after Fig. 6 and Fig. 13. The recorded runs pass exact tests of the rewritten Lemma 7 construction on five fixed instances and 500 random hedges with random liftings checked for graph admissibility (acyclic variable-level graph, disjoint branches); most random instances are small. The revised Def. 1 agrees with independent path enumeration on all ADMGs with at most four vertices; finite checks support the algorithmic and graphical repairs. These checks do not prove the general statements and do not establish completeness.

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AI review sign-off: claude-opus-5-5 / gpt-6.1-sol · 2026-10-25.